INFRASTRUCTURE ONTOLOGY & TAXONOMY

AI Data Centre Facilities Assurance Framework

Version 1.1

Canonical semantics for capability-centred AI infrastructure assurance

Artifact ID: AIDCF-003 (short code IOT)
Artifact type: Normative foundation standard
Status: Draft for controlled review
Date: 2026-10-10
Author: Siva Sethumadhavan
Release 1: Foundational Methodology

This document defines the canonical entity classes, taxonomies, relationships, properties, constraints and machine-readable expectations used to represent AI infrastructure within AIDCF.

Document control

Field	Value
Title	AIDCF Infrastructure Ontology and Taxonomy
Artifact ID	AIDCF-003 (short code IOT)
Version	1.1
Date	2026-10-10
Artifact type	Normative foundation standard
Status	Draft for controlled review
Author	Siva Sethumadhavan
Depends on	AIDCF-001 AIDCF Manifesto; AIDCF-002 Core Conceptual Model
Aligned with	AIDCF-007A Canonical Master Control Registry (15 domains, 90 families, 360 controls); AIDCF-014 End-to-End Traceability and Machine-Executable Data Standard (§7 object classes, §9 relationship types)
Downstream users	AIDCF-004; AIDCF-005; AIDCF-008; AIDCF-009; AIDCF-010; AIDCF-016; AIDCF-018 domain guides; compatible tooling
Normative language	Shall / shall not = requirement; should / should not = recommendation; may = permission; can = capability, as defined in AIDCF-002
Intended audience	AI infrastructure architects; data-centre engineers; assessors; operators; owners; technology teams; resilience, security and assurance practitioners; ontology engineers; product teams
Publication boundary	Independent methodology. Not a substitute for law, licensed engineering, safety approval, regulatory certification or vendor design authority. AIDCF does not certify.

Revision history

Version	Date	Summary
1.0	—	Controlled-review baseline for expert review and machine-readable implementation.
1.1	2026-10-10	Re-baselined taxonomy and class registry to the 15 AIDCF-007A domain codes; added assurance-chain classes (Part XV) and IEC 62443 IACS concepts (Part XVI); adopted the canonical 16-step chain, "Required Function", the `https://w3id.org/aidcf/ontology#` namespace and AIDCF-014 predicate mappings; aligned confidence to CONF-0..CONF-4; public artifact IDs; consecutive requirement IDs (legacy mapping in Appendix G).

Relationship to AIDCF-002

AIDCF-002 Core Conceptual Model defines what the AIDCF concepts mean and how they reason together; this standard defines how those concepts are represented as classes, properties, relationships, controlled values and validation rules. Every class in this ontology realises a concept defined in AIDCF-002, and where a definition here is shorter or more formal, the AIDCF-002 meaning governs. The canonical assurance chain, the definition of MATERIAL, the claim types, result, gate and decision vocabularies, and the legacy-to-15 domain mapping are owned by AIDCF-002 and are reused here without redefinition. AIDCF-014 then fixes the exchange names of the same classes and relationships; this ontology maps to those names explicitly (Appendix A, Appendix B).

Executive thesis

AIDCF represents an AI data centre as a capability-supporting system rather than a collection of equipment. The ontology begins with a defined Capability Instance and Service Outcomes, traces them to infrastructure functions and assets, expresses direct, indirect and shared dependencies, records capacity and operating conditions, and provides stable attachment points for failures, safeguards, evidence and assurance decisions.

Canonical chain

This standard uses the canonical AIDCF assurance chain defined in AIDCF-002 §9 without modification:

1 Decision Question → 2 Capability Instance → 3 Service Outcome → 4 Required Function → 5 Architecture (InfrastructureSystem / AssetInstance / Interface) → 6 Dependency and Fault Domain → 7 Threat / Hazard / Failure Scenario → 8 Safeguard → 9 Control → 10 Evidence Requirement → 11 Evidence Object → 12 Control Assessment (result) → 13 Finding / UNKNOWN Record → 14 Gate Result → 15 Score Snapshot and Decision Recommendation → 16 Assurance Decision → (Assurance Report and Reassessment Trigger close the loop back to 2).

Parts III–XI of this standard specify the classes for steps 2–6 and the attachment points for steps 7–8 (whose semantics are owned by AIDCF-004). Part XV specifies the classes for steps 1 and 9–16 and the closing loop. Operating states (Part X) qualify every step as context and assumptions.

How to use this standard

- Use Parts I-IV to establish a conformant assessment information model.
- Use Parts V-VII to classify infrastructure across the fifteen AIDCF domains (GOV, CAP, DEP, PHY, POW, THM, ENV, CMP, STO, NET, OPS, RES, SUP, SEC, EVA).
- Use Parts VIII-XI to represent relationships, dependencies, states, capacity, responsibility and uncertainty.
- Use Parts XII-XIV to validate records, create implementation profiles and produce machine-readable releases.
- Use Parts XV-XVI to represent the assurance half of the chain (controls, evidence, results, gates, decisions, reports, triggers) and industrial control-system security concepts.
- Do not treat the taxonomy as a control checklist or a substitute for engineering analysis.

Contents

1. Part I - Authority, scope and conformance
2. Part II - Ontology architecture and design principles
3. Part III - Upper ontology and canonical entities
4. Part IV - Capability, outcome, boundary and lifecycle model
5. Part V - Fifteen-domain infrastructure taxonomy
6. Part VI - Cross-cutting classification facets
7. Part VII - Asset, system, interface and resource model
8. Part VIII - Relationship and dependency vocabulary
9. Part IX - Capacity, redundancy, fault-domain and resilience semantics
10. Part X - Operating state, temporal state and change semantics
11. Part XI - Responsibility, evidence, confidence and UNKNOWN semantics
12. Part XII - Conformance rules and validation shapes
13. Part XIII - Reference implementation profiles
14. Part XIV - Machine-readable publication specification
15. Part XV - Assurance-chain classes
16. Part XVI - Industrial automation and control system security concepts (IEC 62443 alignment)
17. Appendix A - Canonical class registry
18. Appendix B - Canonical relationship registry
19. Appendix C - Minimum property schemas
20. Appendix D - Naming, identifier and namespace rules
21. Appendix E - Standards and reference-source map
22. Appendix F - Review and acceptance record
23. Appendix G - Legacy identifier mapping

Part I - Authority, scope and conformance

1. Purpose

This artifact establishes the controlled vocabulary and semantic structure by which AIDCF describes AI capabilities and the physical, engineered, technical, operational, human and external infrastructure on which they depend. It is intended to create consistency across documents, assessments, datasets, knowledge graphs, digital twins and future tooling.

2. Objectives

- Provide a capability-first upper ontology.
- Define a vendor-neutral taxonomy organised by the fifteen AIDCF domain codes of AIDCF-007A.
- Express cross-domain and external dependencies without hiding scope boundaries.
- Distinguish systems, assets, components, interfaces, resources, services, roles and evidence.
- Represent capacity, redundancy, fault domains, operating states and temporal change.
- Preserve provenance, confidence, conflict and UNKNOWN.
- Provide machine-readable semantics without requiring proprietary software.
- Provide the classes on which failure analysis (AIDCF-004), control mapping (AIDCF-007A), evidence (AIDCF-008), assessment (AIDCF-009), gates and decisions (AIDCF-010) and reporting (AIDCF-012) attach.

3. Scope

The ontology covers entities materially relevant to the ability of a defined AI Capability Instance to deliver specified Service Outcomes. It spans the fifteen AIDCF domains: governance (GOV); capability, outcome and demand (CAP); dependency and external reliance (DEP); physical site, building and security (PHY); electrical power (POW); cooling, thermal and fluid systems (THM); environmental and resource performance (ENV); AI compute (CMP); AI storage and data (STO); AI network and interconnect (NET); building operations (OPS); resilience and recovery (RES); supply chain (SUP); digital, OT and control-plane security (SEC); and evidence, assessment and assurance management (EVA).

4. Exclusions

- Model quality, fairness, explainability and responsible-AI governance, except where a digital element is an infrastructure dependency.
- Application business logic, agent authority and model-to-tool trust paths, which remain primarily within AI Trust Graph.
- Legal compliance determinations and jurisdiction-specific certification decisions.
- Professional engineering approval, safety certification and life-safety design authority.
- Vendor endorsement or prescriptive selection of accelerator, fabric, storage or cooling technology.

5. Normative conformance

> **IOT-REQ-001** A conformant implementation shall preserve the canonical meanings and required distinctions in this artifact.
>
> **IOT-REQ-002** Extensions may add classes, relationships or properties but shall not silently redefine canonical terms.
>
> **IOT-REQ-003** Every material entity shall possess a stable identifier, class, lifecycle status, provenance, evidence status and, where it supports an assurance claim, a confidence value (CONF-0 to CONF-4 per AIDCF-008).
>
> **IOT-REQ-004** Every assessment dataset shall identify the Capability Instance, Assessment Boundary, applicable Service Outcomes and Operating Envelope.
>
> **IOT-REQ-005** Dependencies crossing the Assessment Boundary shall remain represented as external dependencies or UNKNOWN; they shall not disappear because they are out of scope.
>
> **IOT-REQ-006** UNKNOWN shall remain a first-class semantic state and shall not be converted into pass, fail, zero, not applicable or accepted risk.

6. Conformance levels

Level	Name	Minimum expectation
IOT-CL0	Vocabulary alignment	Uses canonical terms and the fifteen domain codes without contradictory meanings.
IOT-CL1	Structured representation	Implements required entities, identifiers and core relationships.
IOT-CL2	Assurance-ready dataset	Adds boundaries, outcomes, dependencies, states, evidence, confidence, the Part XV assurance classes and validation.
IOT-CL3	Graph-conformant implementation	Implements machine-readable schema, AIDCF-014 relationship mappings and validation shapes.
IOT-CL4	Controlled extension	Publishes namespace, mappings, governance, tests and change impact for extensions.

Conformance levels IOT-CL0 to IOT-CL4 were labelled C0 to C4 in v1.0; they are renamed so that they cannot be confused with the evidence coverage dimension C of AIDCF-008.

Part II - Ontology architecture and design principles

7. Four-layer semantic architecture

Layer	Purpose	Examples
L0 Doctrine	Constrains meaning and reasoning	Capability-first; evidence-gated; cross-domain; UNKNOWN preservation (AIDCF-001, AIDCF-002).

Layer	Purpose	Examples
L1 Upper ontology	Defines universal entity types	Capability, Outcome, System, Asset, Dependency, State, Actor, Evidence.
L2 Domain taxonomy	Classifies domain-specific infrastructure	UPS, CDU, accelerator node, parallel file system, scale-out fabric.
L3 Profiles and extensions	Specialises for deployment patterns	Sovereign AI, colocation, liquid-cooled training, enterprise inference.

8. Design principles

> **P1 Capability primacy** The ontology begins with what the infrastructure must enable.
>
> **P2 Instance specificity** An assessed deployment is represented as an instance, not merely a generic type.
>
> **P3 Function before product** Required functions are separated from the products that implement them.
>
> **P4 Relationship explicitness** Material dependencies and interfaces are encoded, not left only in narrative.
>
> **P5 State awareness** Claims are tied to declared operating states (NORMAL, DEGRADED, MAINTENANCE, EMERGENCY, RECOVERY, TRANSITION, SHUTDOWN, FAILED, UNKNOWN; §45).
>
> **P6 Time awareness** Design, as-built, commissioned and current states are distinguishable.
>
> **P7 Evidence anchoring** Material assertions can be linked to evidence and provenance.
>
> **P8 Vendor neutrality** Canonical classes do not require one supplier, topology or technology.
>
> **P9 Safe extensibility** Profiles can specialise the model without breaking semantic compatibility.
>
> **P10 Minimal false precision** Unknown or conflicting information remains visible.

9. Metamodel

Each ontology term is described through a common metamodel.

Field	Meaning
Canonical ID	Stable AIDCF identifier.
Preferred label	Authoritative human-readable name.
Definition	Necessary and distinguishing meaning.
Entity type	Class, relationship, property, enumeration or constraint.
Parent	Immediate broader concept.
Domain	One primary AIDCF domain code (GOV … EVA) and optional secondary codes, or Upper for cross-domain upper-ontology classes.
Allowed relations	Permitted outgoing and incoming relationships.
Required properties	Minimum data needed for conformance.
Lifecycle status	Draft, released, deprecated or retired.
Provenance	Origin, author, approval and version.
Examples	Non-normative illustrations.
Exclusions	Concepts intentionally not covered by the term.

Part III - Upper ontology and canonical entities

10. Root entities

Class ID	Definition
aidcf:Entity	Root of all addressable objects.
aidcf:CapabilityEntity	An intended ability and its bounded implementation.
aidcf:OutcomeEntity	A required or observed result.
aidcf:InfrastructureEntity	A system, asset, component, interface, location or resource.
aidcf:ConditionEntity	A state, constraint, assumption, limit or environmental condition.
aidcf:AdverseEntity	A threat, hazard, initiating event, cause, failure mode or consequence.
aidcf:AssuranceEntity	A safeguard, control, assertion, evidence, test, finding or decision.
aidcf:PartyEntity	An organisation, actor, role, team or supplier.
aidcf:TemporalEntity	A lifecycle stage, timestamp, validity interval or change event.

11. Canonical class definitions

Preferred term	Canonical definition
Capability	An intended operational ability supported by infrastructure.

Preferred term	Canonical definition
Capability Instance	A specific deployment bounded by ownership, location, architecture, capacity, criticality and obligations.
Service Outcome	A measurable or assessable result required from a Capability Instance.
Required Function	A function whose loss or degradation would materially affect a Service Outcome (formerly "Required Function").
Assessment Boundary	The declared inclusion perimeter for direct assessment.
Operating Envelope	The bounded conditions under which a capability or system is intended to operate.
Infrastructure System	A coordinated set of assets and interfaces providing one or more functions.
Asset Instance	An identifiable real-world item, service, person, procedure or resource.
Component	A constituent part of an asset or system represented where material.
Interface	A defined point or mechanism of exchange between entities.
Connection	A realised physical, logical, hydraulic, electrical, thermal or organisational link.
Resource	Consumable or constrained input required for operation.
Location	A geographic, site, building, zone, room, row, rack or logical placement.
Dependency	A reliance by one entity upon another entity, resource, condition or decision.
Fault Domain	A set of entities expected to be affected by a common failure.
Redundancy Group	Assets or paths deliberately arranged to preserve a function.
Safeguard	A real-world measure intended to anticipate, prevent, detect, contain, respond, recover or adapt.
Evidence Object	A record or observation used to support, challenge or qualify an assertion (Part XV).
Actor	A human or machine participant with a relevant role.
Organisation	A legally or operationally recognised party.
Role	A named set of responsibilities and authorities.
Operating State	A defined condition of operation.
Change Event	An approved, observed or unapproved modification affecting the represented system.

12. Mandatory distinctions

Do not collapse	Reason
Capability / Capability Instance	A generic ability is not the assessed deployment.
System / Asset / Component	Composition and assessment granularity differ.
Function / Implementing asset	A function can be reallocated or implemented by multiple technologies.
Interface / Dependency	An interface enables exchange; a dependency is reliance.
Connection / Relationship	A connection is realised; a relationship may be logical or asserted.
Capacity / Demand / Reserve	They answer different questions.
Redundancy / Diversity / Independence / Resilience	Additional paths do not prove common-cause independence or outcome resilience.
Designed / Installed / Commissioned / Observed state	Documentation and reality may diverge.
Owner / Operator / Maintainer / Risk owner	Commercial and operational responsibilities differ.
UNKNOWN / NOT_APPLICABLE / NONCONFORMING / NOT_ASSESSED	Evidence insufficiency is not the same as approved non-applicability, demonstrated failure or absence of assessment.
Safeguard / Control	A safeguard is the real-world measure; a control is the assessable requirement in AIDCF-007A evaluated against it.
Evidence status / Confidence	Evidence status describes a record's support (§53); confidence (CONF-0 to CONF-4) is the assessor's bounded belief in a claim.

Part IV - Capability, outcome, boundary and lifecycle model

13. Capability hierarchy

Level	Entity	Illustration
1	Capability family	AI training; inference; fine-tuning; HPC simulation; shared accelerator service.
2	Capability type	Large-scale distributed training; low-latency inference; batch inference.
3	Capability Instance	A named deployment at a defined site with a declared architecture.
4	Service Outcome	Availability, throughput, completion, latency, integrity or recovery outcome.
5	Required Function	Heat removal, power continuity, scheduling, checkpoint persistence, fabric transport.
6	Supporting system and assets	Cooling loop, UPS path, compute cluster, storage system, network fabric.

14. Capability Instance minimum record

Property	Requirement
capabilityInstanceId	Globally or organisationally unique and immutable.
name	Human-readable identifier.

Property	Requirement
capabilityType	Controlled vocabulary value.
owner	Accountable organisation or role.
locations	Sites and zones in which it is instantiated.
lifecycleStage	PLAN, DESIGN, BUILD, COMMISSION, OPERATE, CHANGE or DECOMMISSION (§17).
serviceOutcomes	One or more outcome records.
criticality	Declared and justified criticality.
assessmentBoundary	Included systems and explicit exclusions.
operatingEnvelope	Load, environmental, capacity, duration and state constraints.
architectureBaseline	Approved design or configuration reference.
effectivePeriod	Start/end or point-in-time applicability.
evidenceStatus	CONFIRMED, PARTIALLY_CONFIRMED, ASSERTED, INFERRED, CONFLICTING, STALE or UNKNOWN (§53).

15. Service Outcome taxonomy

Outcome family	Examples of assessable properties
Availability	Required service continuity; tolerated interruption; degraded-mode availability.
Capacity	Minimum usable accelerator, storage, network, power and cooling capacity.
Performance	Throughput, latency, job completion, checkpoint duration, fabric efficiency.
Integrity	Correct configuration, artifact integrity, data-path integrity and state consistency.
Isolation	Tenant, cluster, management, physical and fault-domain separation.
Recoverability	Recovery objective, restart conditions, restoration sequence and validation.
Safety	Safe thermal, electrical, fire, structural and human operating conditions.
Efficiency	Resource performance within defined resilience constraints.
Sovereignty	Declared location, operational control, supply, maintenance and dependency constraints.
Observability	Minimum telemetry, alarm and diagnostic visibility needed to operate and assure.

16. Boundary model

> **IOT-REQ-007** An Assessment Boundary shall identify included Capability Instances, sites, zones, systems, assets, interfaces, lifecycle stages and time horizon.
>
> **IOT-REQ-008** Every exclusion shall identify rationale, owner, residual dependency and effect on confidence.
>
> **IOT-REQ-009** External services shall be represented through boundary-crossing relationships and shall not be treated as inherently reliable.

17. Lifecycle stages

Code	Stage	Ontology emphasis
PLAN	Concept	Capability intent, candidate site, constraints, assumptions and target outcomes.
DESIGN	Design	Required Functions, systems, architecture, capacity, fault domains and interfaces.
BUILD	Construction	Asset instances, installation state, deviations and provenance.
COMMISSION	Commissioning and operational readiness	Tests, observations, results, outstanding defects, accepted limitations; procedures, personnel, spares, monitoring and transfer of authority.
OPERATE	Operation	Current state, telemetry, maintenance, incidents, evidence and drift.
CHANGE	Change	Impact, configuration transition, temporary states, evidence and revalidation.
DECOMMISSION	Decommissioning	Isolation, removal, sanitisation, residual hazards and record retention.

The codes are the canonical lifecycle stages of AIDCF-002. v1.0 listed operational readiness as a separate stage; it is the closing part of COMMISSION.

Part V - Fifteen-domain infrastructure taxonomy

18. Taxonomy rules

> **IOT-REQ-010** Domain assignment shall indicate the primary classification only; cross-domain relationships shall remain explicit.
>
> **IOT-REQ-011** Every class in Appendix A shall carry exactly one primary domain code from the fifteen AIDCF domain codes of AIDCF-007A, or the tag Upper for cross-domain upper-ontology classes, and may carry secondary codes where controls in another domain routinely assess the class.
>
> **IOT-REQ-012** A domain code on a class shall not be read as restricting which AIDCF-007A controls may reference the class; controls declare the classes they assess through their ontologyClasses attribute.
>
> **IOT-REQ-013** Technology-specific subclasses shall inherit canonical properties from vendor-neutral parent classes.

> **IOT-REQ-014** A taxonomy branch shall be extended only where the distinction changes assessment, dependency, evidence, failure or assurance meaning.

19. Domain index

The taxonomy uses the fifteen domain codes of AIDCF-007A. The superseded eight-domain model is mapped to these codes in AIDCF-002 Part X; it is not used in this standard.

Code	Domain	AIDCF-007A families	Described in
GOV	Governance, Accountability and Assurance Direction	GOV-01 to GOV-06 (MCF-GOV-001 to MCF-GOV-024)	§20.1
CAP	AI Capability, Service Outcome and Demand Management	CAP-01 to CAP-06 (MCF-CAP-001 to MCF-CAP-024)	§20.2
DEP	Dependency, Concentration and External Reliance Management	DEP-01 to DEP-06 (MCF-DEP-001 to MCF-DEP-024)	§20.3
PHY	Physical Site, Building and Security Infrastructure	PHY-01 to PHY-06 (MCF-PHY-001 to MCF-PHY-024)	§21
POW	Electrical Power Infrastructure and Energy Continuity	POW-01 to POW-06 (MCF-POW-001 to MCF-POW-024)	§22
THM	Cooling, Thermal and Fluid Infrastructure	THM-01 to THM-06 (MCF-THM-001 to MCF-THM-024)	§23
ENV	Environmental and Resource Performance	ENV-01 to ENV-06 (MCF-ENV-001 to MCF-ENV-024)	§24
CMP	AI Compute Infrastructure	CMP-01 to CMP-06 (MCF-CMP-001 to MCF-CMP-024)	§25.1
STO	AI Storage and Data Infrastructure	STO-01 to STO-06 (MCF-STO-001 to MCF-STO-024)	§25.2
NET	AI Network and Interconnect Architecture	NET-01 to NET-06 (MCF-NET-001 to MCF-NET-024)	§25.3
OPS	Building Operations and Service Management	OPS-01 to OPS-06 (MCF-OPS-001 to MCF-OPS-024)	§26.1
RES	Resilience, Continuity and Recovery	RES-01 to RES-06 (MCF-RES-001 to MCF-RES-024)	§26.2
SUP	Supply Chain, OEM and Ecosystem Assurance	SUP-01 to SUP-06 (MCF-SUP-001 to MCF-SUP-024)	§26.3
SEC	Digital, OT and Control-Plane Security	SEC-01 to SEC-06 (MCF-SEC-001 to MCF-SEC-024)	§26.4
EVA	Evidence, Assessment and Assurance Management	EVA-01 to EVA-06 (MCF-EVA-001 to MCF-EVA-024)	§26.5

20. Direction and definition domains (GOV, CAP, DEP)

20.1 GOV Governance, Accountability and Assurance Direction

GOV classes represent the authority structure behind assurance: who may approve, delegate, accept risk and grant exceptions. They are the objects to which AIDCF-007A GOV-01 to GOV-06 controls and gate CG-13 attach. Individual roles and organisations use the upper classes Role and Organisation (§11).

Taxonomy family	Illustrative subclasses
Governance and authority	Operating Model, Delegation, Approval Authority, Escalation Path, Risk Acceptance
Obligations and baselines	Policy, Technical Standard Baseline, Exception Record, Contractual Obligation

20.2 CAP AI Capability, Service Outcome and Demand Management

CAP classes hold the capability, demand and engineering-baseline records against which every other domain is assessed. The Capability Instance, Service Outcome, Required Function and Operating Envelope themselves are upper classes (§11, Part IV) tagged CAP in Appendix A. Engineering records sit here because they define design intent (CAP-06 design-to-operation traceability); each also carries the code of the engineering domain it documents.

Taxonomy family	Illustrative subclasses
Capability and demand records	Capability Register, Demand Profile, Load Step Profile, Growth Forecast
Engineering baseline and records	Design Basis, Single-Line Diagram, P&ID, As-Built Drawing, Engineering Study, Commissioning Script, Deviation Record

20.3 DEP Dependency, Concentration and External Reliance Management

DEP classes make reliance across the assessment boundary representable. The Dependency and Assessment Boundary classes are upper classes (§11, Part VIII); the classes below represent the external services and boundary crossings that DEP-03 and DEP-05 controls examine.

Taxonomy family	Illustrative subclasses
Boundary and external service	External Service, Boundary Crossing, Service Interface Agreement

21. PHY Physical Site, Building and Security Infrastructure

PHY covers the site, the building fabric and structure that carries AI equipment, physical security, and fire and life-safety systems. Civil and structural and fire classes were classified under facilities engineering in v1.0.

Taxonomy family	Illustrative subclasses
Site and external perimeter	Site, Campus, Property Boundary, Perimeter Barrier, Vehicle Barrier, Gate, Guard Post, External Approach, Standoff Zone
Building and internal zoning	Building, Floor, Data Hall, Room, Cage, Suite, Row, Rack, Cabinet, Security Zone, Restricted Area, Loading Area, Staging Area
Civil and structural	Foundation, Floor Slab, Raised Floor, Roof, Wall, Ceiling, Rack Anchorage, Seismic Restraint, Load-Bearing Element
Entry and identity control	Access Point, Turnstile, Mantrap, Door, Lock, Credential Reader, Biometric Reader, Visitor System, Key System
Detection and surveillance	Camera, Video Management System, Intrusion Sensor, Door Contact, Motion Sensor, Security Alarm, Intercom
Asset movement and custody	Loading Bay, Goods Lift, Screening Point, Storage Area, Seal, Chain-of-Custody Record

Taxonomy family	Illustrative subclasses
Physical response	Security Operations Centre, Guard Force, Response Procedure, Dispatch Channel, Incident Reconstruction Record
Fire and life safety	Detection Zone, Fire Alarm Panel, Suppression Zone, Suppression Agent System, Sprinkler System, Smoke Control, Emergency Egress, Emergency Power Off

22. POW Electrical Power Infrastructure and Energy Continuity

POW covers the electrical path from utility connection to the rack power shelf, including generation, energy storage, protection and electrical monitoring.

Taxonomy family	Illustrative subclasses
Utility and grid interface	Incoming Supply, Utility Feed, Grid Connection, Substation
Generation and fuel	On-Site Generation, Generator, Fuel Tank, Fuel System, Starting System, Exhaust System
Conversion, energy storage and distribution	Transformer, Switchgear, MV Switchgear, LV Switchgear, Transfer Scheme, Static Transfer Switch, UPS, Battery, Energy Storage, Busway, PDU, RPP, Rack PDU, Power Supply Unit
Protection, earthing and power quality	Protection Relay, Protection Device, Relay, Earthing System, Grounding, Lightning Protection, Harmonic Filter, Surge Protection
Electrical monitoring	EPMS, Meter, Power Quality Monitor

23. THM Cooling, Thermal and Fluid Infrastructure

THM covers the heat path from the accelerator package to heat rejection: heat capture, coolant distribution, facility water, air-side systems, fluid quality and leak management.

Taxonomy family	Illustrative subclasses
Air-side and mechanical systems	Chilled Water Plant, Condenser System, Pump System, Air Distribution, Ventilation, Pressurisation, Air Handler, In-Row Cooler, Rear-Door Heat Exchanger
Heat capture at the IT load	Heat Sink, Cold Plate, Immersion Tank
Coolant distribution	CDU, Primary Loop, Secondary Loop, Pump, Manifold, Quick Disconnect, Hose, Pipe, Heat Exchanger
Heat rejection	Chiller, Cooling Tower, Dry Cooler, Condenser, Evaporative System, Heat-Reuse Interface
Fluid, water and leak management	Water Source, Pipework, Valve, Drain, Sump, Drainage, Leak Detection Zone, Water Treatment System
Thermal and leak monitoring	Temperature Sensor, Humidity Sensor, Flow Sensor, Pressure Sensor, Leak Sensor, Coolant Quality Sensor, Thermal Camera

24. ENV Environmental and Resource Performance

ENV classes are resources, conditions, indicators and obligations rather than equipment. They are used to qualify capacity assertions (§41), operating envelopes and environmental claims.

Taxonomy family	Illustrative subclasses
Energy resources	Grid Electricity, Renewable Electricity, On-Site Energy, Stored Energy, Fuel
Water resources	Potable Water, Process Water, Reclaimed Water, Cooling Water, Discharge Water
Environmental conditions	Ambient Temperature, Humidity, Air Quality, Particulate Level, Noise, Vibration, Flood Exposure, Heat Stress
Material and chemical resources	Refrigerant, Coolant, Fire Suppressant, Battery Chemistry, Hazardous Material
Performance indicators	PUE Record, WUE Record, CUE Record, Energy Reuse Record, Cooling Efficiency Record, Compute Utilisation Record
Environmental obligations	Permit, Reporting Requirement, Emission Limit, Water Limit, Waste Requirement, Environmental Commitment

25. AI infrastructure domains (CMP, STO, NET)

25.1 CMP AI Compute Infrastructure

CMP covers accelerator hardware, rack-scale systems, clusters and the firmware and management software that make them usable. Vendor-specific subclasses (for example NVLink Domain, Kubernetes Cluster) are specialisations of their vendor-neutral parents (Scale-Up Domain, Orchestrator) per IOT-REQ-013.

Taxonomy family	Illustrative subclasses
Accelerated compute	Accelerator, GPU, AI ASIC, Compute Node, GPU Baseboard, CPU, Memory, DPU, NIC
Rack-scale systems	Compute Tray, Rack-Scale System, Scale-Up Domain, NVLink Domain, Enclosure, Backplane, Power Shelf, Rack Controller
Cluster systems	Compute Cluster, Cluster Partition, Node Pool, Head Node, Login Node, Management Node, Provisioning Node
Management and firmware	BMC, Firmware, BIOS, Driver, Cluster Manager, Provisioning Service, Scheduler, Orchestrator
Workload infrastructure	Queue, Scheduler Partition, Reservation, Container Runtime, Kubernetes Cluster, Job Scheduler, Checkpoint Agent
Compute observability	GPU Telemetry, Node Telemetry, Health Service, Performance Counter, Error Counter
Compute lifecycle	Golden Image, Firmware Baseline, Compatibility Matrix, Hardware Support Contract, RMA Process, Secure Decommissioning Record

25.2 STO AI Storage and Data Infrastructure

STO covers storage systems, data-path services and the repositories that hold datasets, models and checkpoints, including protection copies used for recovery.

Taxonomy family	Illustrative subclasses
Storage systems	Parallel File System, Scale-Out NAS, Object Storage, Block Storage, Local NVMe, Burst Buffer, Archive Storage
Storage nodes and media	Storage Node, Metadata Server, Object Storage Node, Controller, SSD, HDD, Tape, NVRAM

Taxonomy family	Illustrative subclasses
Data-path services	Namespace, Metadata Service, Data Mover, Cache, Data-Path Gateway, Client Driver, Mount Service
AI artifact services	Dataset Repository, Model Repository, Checkpoint Repository, Feature Store, Artifact Registry
Protection and recovery	Snapshot, Replica, Backup Copy, Erasure Coding Group, Recovery Copy, Immutable Copy
Storage management	Storage Manager, Capacity Manager, Tiering Policy, Retention Policy, Integrity Scanner, Telemetry Service
Data logistics interfaces	Ingestion Interface, Export Interface, Data Transfer Node, WAN Transfer Service, Data Staging Area

25.3 NET AI Network and Interconnect Architecture

NET covers scale-up and scale-out fabrics, storage, frontend and management networks, core network services and physical connectivity.

Taxonomy family	Illustrative subclasses
Scale-up interconnect	Scale-Up Fabric, Switch Fabric, Interconnect Link, Topology Domain
Scale-out compute fabric	InfiniBand Fabric, Ethernet AI Fabric, RoCE Fabric, Compute Leaf, Compute Spine, Fabric Manager
Storage fabric	Storage Network, Storage Leaf, Storage Spine, SAN Fabric, Storage Gateway
Frontend and service network	Frontend Network, Load Balancer, Service Gateway, Edge Router, Firewall, API Network
Management and control network	Management Network, Out-of-Band Network, Console Server, Management Switch, Jump Service
Timing and core services	Time Service, DNS Service, DHCP Service, IPAM Service, Certificate Service, Authentication Service
Physical connectivity	Cable, Optical Fibre, Copper Link, Transceiver, Patch Panel, Meet-Me Room, Cross-Connect
Network observability	Flow Telemetry, Fabric Telemetry, Packet Capture Service, Network Controller, Congestion Monitor, Path Monitor

26. Operation, resilience and assurance domains (OPS, RES, SUP, SEC, EVA)

26.1 OPS Building Operations and Service Management

OPS covers how the facility is run: work control, maintenance, staffing, alarms and incidents, configuration records, and the facility monitoring and control systems operated day to day. Facility control systems are also industrial automation and control systems in the sense of Part XVI and carry SEC as a secondary code.

Taxonomy family	Illustrative subclasses
Work control	Procedure, Method Statement, Permit to Work, Lockout Tagout, Maintenance Window, Change Record, Temporary Configuration
Maintenance and reliability	Maintenance Plan, Work Order, Condition Monitoring, Inspection, Calibration, Spare Part, Obsolescence Record
Staffing and competence	Shift, Roster, Competency Record, Authorisation, Training Record, Contractor, Subject Matter Expert
Alarm and incident	Alarm, Incident, Event Log, Post-Incident Review
Configuration and documentation	Configuration Baseline, Asset Register, Drawing Register, Procedure Register, Revision, Document Custodian
Facility monitoring and control	BMS, SCADA, PLC, RTU, Sensor, Actuator, Control-System Gateway, Historian, Alarm Console

26.2 RES Resilience, Continuity and Recovery

RES classes represent continuity and recovery arrangements. Fault Domain and Redundancy Group are upper classes (§11, Part IX) tagged RES; protection copies in STO and golden images in CMP carry RES as a secondary code.

Taxonomy family	Illustrative subclasses
Continuity and emergency response	Emergency Procedure, Crisis Team, Exercise, Continuity Plan, Recovery Plan, Alternative Capacity

26.3 SUP Supply Chain, OEM and Ecosystem Assurance

SUP covers supplier relationships, service obligations and the provenance of hardware, software and firmware. Supplier management classes were classified under building operations in v1.0.

Taxonomy family	Illustrative subclasses
Supplier and service management	Supplier, Maintenance Contract, Service Level, Remote Support, Escalation Contact, Lead Time, Supply Constraint
Provenance and software supply	Software Release, Signature Record, Software Bill of Materials, Component Provenance Record

26.4 SEC Digital, OT and Control-Plane Security

SEC covers identity, privileged and remote access, security monitoring, and authority for automated or AI-enabled action over infrastructure. Industrial control-system zones, conduits and security levels are specified in Part XVI.

Taxonomy family	Illustrative subclasses
Identity and privileged access	Identity Provider, Human Identity, Machine Identity, Privileged Access Service, Break-Glass Account, Remote Access Gateway
Security monitoring and response	Security Telemetry Service, Security Incident, Vulnerability Record
Automated and AI-enabled action	Automation Agent, Command Allowlist, Approval Workflow

26.5 EVA Evidence, Assessment and Assurance Management

EVA has no infrastructure taxonomy of its own: its classes are the assurance-chain classes of Part XV (Control Domain to Reassessment Trigger), tagged EVA in Appendix A.

Part VI - Cross-cutting classification facets

27. Facet model

Domain classification alone is insufficient. Every material entity may also be classified through orthogonal facets that support analysis without creating duplicate classes.

Facet	Controlled values / meaning
Criticality	Critical, High, Moderate, Low, Unclassified, UNKNOWN.
Lifecycle status	Planned, designed, procured, installed, commissioned, operational, standby, degraded, under maintenance, bypassed, failed, recovering, retired.
Physicality	Physical, logical, human, procedural, contractual, environmental, informational.
Boundary position	Internal, boundary-crossing, external, excluded-but-dependent, UNKNOWN.
Tenancy	Dedicated, shared, multi-tenant, pooled, reserved, burst, UNKNOWN.
Control authority	Direct, delegated, contractual, advisory, none, UNKNOWN.
Evidence status	CONFIRMED, PARTIALLY_CONFIRMED, ASSERTED, INFERRED, CONFLICTING, STALE, UNKNOWN (§53).
Replaceability	Immediate, stocked spare, contractually replaceable, long-lead, irreplaceable, UNKNOWN.
Mobility	Fixed, relocatable, portable, virtualised, service-based.
Exposure	Public, campus, building, hall, cage, rack, management plane, isolated, UNKNOWN.
Sovereignty relevance	Local control, foreign dependency, cross-border support, remote administration, jurisdiction-sensitive, not declared.
Environmental sensitivity	Temperature, humidity, water quality, vibration, particulate, altitude, electromagnetic, acoustic.

28. Criticality semantics

> **IOT-REQ-015** Criticality shall be assigned in relation to a Capability Instance and Service Outcome, not as an intrinsic universal property of an asset class.
>
> **IOT-REQ-016** Criticality rationale should consider consequence, time sensitivity, substitutability, concentration, recovery complexity, safety, regulatory and strategic significance.

29. Classification inheritance

A subclass inherits properties and permitted relationships from its parent unless an approved extension explicitly narrows or adds semantics. Inheritance shall not be used to infer an operational fact about an instance. For example, classifying an asset as a UPS does not establish its capacity, redundancy, condition or effectiveness.

Part VII - Asset, system, interface and resource model

30. System decomposition

Construct	Use
Infrastructure System	Group assets that collectively provide a function.
Subsystem	Represent a meaningful functional subdivision.
Asset Group	Group similar instances for inventory or analysis.
Functional Group	Group heterogeneous assets that realise a function.
Redundancy Group	Represent alternatives designed to preserve a function.
Fault Domain	Represent common failure exposure.
Protection Zone	Represent a security, fire, electrical or thermal boundary.
Maintenance Group	Represent assets taken out of service together.
Capacity Pool	Represent shared capacity available to consumers.

31. Interface taxonomy

Interface class	Examples	Minimum properties
Electrical	Supply, feeder, bus, breaker termination	Voltage, phase, capacity, protection, source, destination.
Thermal	Air or liquid heat transfer point	Medium, temperature, flow, pressure, capacity, direction.
Hydraulic	Coolant or water connection	Fluid, pressure, flow, valve/isolation, leak boundary.
Network	Port, link, VLAN, subnet, fabric endpoint	Protocol, bandwidth, topology role, path, security zone.
Mechanical	Mounting, anchoring, ventilation or structural contact	Load, tolerance, connection, isolation.
Control	Sensor, command, alarm or automation interface	Signal, authority, fail state, protocol, authentication.
Human/procedural	Handover, approval, escalation or work interface	Role, trigger, response objective, record.
Contractual/service	Utility, supplier or maintenance service boundary	Provider, service level, capacity, escalation, evidence.

32. Resource taxonomy

Resource	Key semantics
Electrical energy	Source, quality, capacity, duration, carbon attribute, dependency.
Fuel	Type, quantity, quality, replenishment, usable duration.
Cooling capacity	Heat-removal capacity under defined conditions.
Water	Source, quality, quantity, permitted use and continuity.
Coolant/refrigerant	Type, compatibility, quantity, quality and handling constraints.
Space and structural capacity	Area, volume, floor loading, rack loading and access.
Network bandwidth	Usable throughput, latency, oversubscription and state.
Storage capacity	Usable capacity, performance tier, protection overhead and headroom.
Compute capacity	Usable accelerator/CPU/memory capacity and allocation.
Human capacity	Competence, availability, authorisation and concentration.
Time	Transfer time, autonomy, response time, maintenance window, recovery time.
Spares	Quantity, compatibility, location, lead time and preservation state.

33. Minimum asset record

Property	Cardinality	Meaning
assetId	Required	Stable identifier.
assetClass	Required	Canonical or approved extension class.
name	Required	Human-readable name.
systemMembership	Required where applicable	Parent system(s).
location	Required for physical assets	Placement or service jurisdiction.
owner	Required	Accountable owner or UNKNOWN.
operator	Required for operational assets	Operator or UNKNOWN.
maintainer	Conditional	Maintenance party.
lifecycleStatus	Required	Controlled value.
criticality	Required if material	Capability-specific value and rationale.
capacityProperties	Conditional	Units and operating conditions.
interfaces	Conditional	Connections and service boundaries.
dependencies	Required if material	Direct and known indirect reliances.
faultDomains	Required if resilience-relevant	One or more fault-domain memberships.
evidenceLinks	Required for assurance use	Evidence supporting material attributes.
validity	Required	Effective period and observation date.
evidenceStatus	Required	CONFIRMED through UNKNOWN (§53).

Part VIII - Relationship and dependency vocabulary

34. Relationship design rules

> **IOT-REQ-017** Every relationship shall identify subject, predicate and object; material relationships shall also identify scope, conditions, provenance, confidence and effective period.
>
> **IOT-REQ-018** Inverse relationships may be generated only where the canonical registry declares an inverse.
>
> **IOT-REQ-019** Symmetric relationships shall not be assumed for directional dependencies.
>
> **IOT-REQ-020** Transitive reasoning shall be applied only to approved predicates and shall not automatically infer engineering sufficiency.
>
> **IOT-REQ-021** Records exchanged between tools shall use the AIDCF-014 §9 relationship types; ontology predicates in this standard that are not AIDCF-014 types shall be exchanged through the mapping in Appendix B and shall not be emitted as new exchange types.

35. Core structural relationships

Predicate	Subject	Object	Logic	Inverse	Meaning
PART_OF	Entity	Entity	Transitive	HAS_PART	Composition.
CONTAINS	Location/System	Entity	Transitive	LOCATED_IN/PART_OF	Containment.
LOCATED_IN	Physical Entity	Location	Transitive	CONTAINS	Physical placement.
MEMBER_OF	Entity	Group	No	HAS_MEMBER	Group membership.
HAS_OUTCOME	CapabilityInstance	ServiceOutcome	No	—	Capability requires the outcome.

Predicate	Subject	Object	Logic	Inverse	Meaning
REQUIRES_FUNCTION	ServiceOutcome	RequiredFunction	No	—	Outcome depends on the function.
IMPLEMENTED_BY	RequiredFunction	System/Asset	No	IMPLEMENTS	Function is realised by the system or asset.
INTERFACES_WITH	Entity	Entity	Symmetric	INTERFACES_WITH	Defined exchange boundary.
CONNECTS_TO	Entity	Entity	Symmetric unless directed	CONNECTS_TO	Realised connection.
ROUTES_THROUGH	Flow/Service	Entity	No	CARRIES	Path relationship.

v1.0 used SUPPORTS for "entity contributes to capability or outcome". That use is withdrawn: SUPPORTS is reserved for evidence supporting an assertion (AIDCF-014 §9). Contribution to an outcome is expressed as HAS_OUTCOME → REQUIRES_FUNCTION → IMPLEMENTED_BY. IMPLEMENTS is retained only as the inverse of IMPLEMENTED_BY, and CONNECTED_TO is renamed CONNECTS_TO.

36. Functional dependency relationships

Predicate	Subject	Object	Meaning
DEPENDS_ON	Consumer entity	Provider/entity/condition	General reliance.
POWERED_BY	Load/system	Electrical source/path	Electrical reliance.
COOLED_BY	Heat source/system	Cooling system/path	Thermal reliance.
SUPPLIED_BY	Consumer	Provider/resource	Resource or service supply.
CONTROLLED_BY	Asset/system	Controller/actor	Operational or automated control.
MONITORED_BY	Entity	Monitoring entity	Visibility and detection reliance.
PROTECTED_BY	Entity	Safeguard	Protective relationship.
AUTHENTICATED_BY	Entity/service	Identity service	Identity reliance.
TIMED_BY	Entity/service	Time source	Timing reliance.
CONFIGURED_BY	Entity	Configuration source/actor	Configuration reliance.
RECOVERS_THROUGH	Entity/capability	Recovery asset/process	Restoration reliance.
MAINTAINED_BY	Entity	Actor/organisation	Maintenance reliance.
OPERATED_BY	Entity/system	Actor/organisation	Operational responsibility.
EVIDENCED_BY	Assertion/entity	Evidence object	Evidence support.
TESTED_BY	Assertion/safeguard	Test record	Validation link (exchanged as ASSESSED_BY with method TEST).

37. Dependency record schema

Field	Required	Description
dependencyId	Yes	Stable relationship identifier.
consumer	Yes	Entity reliant on another.
provider	Yes or UNKNOWN	Entity, resource or condition relied upon.
dependencyType	Yes	Electrical, thermal, hydraulic, network, control, human, supplier, etc.
strength	Yes	Absolute, material, conditional, supporting, recovery-only or UNKNOWN.
direction	Yes	Provider to consumer effect direction.
applicableStates	Yes	Operating states in which reliance is material.
capacityRequirement	Conditional	Required quantity, unit, duration and quality.
substitutability	Conditional	Alternative provider or workaround.
transferMechanism	Conditional	Automatic, manual, procedural or none.
commonDependencies	Conditional	Shared providers or causes.
owner	Yes or UNKNOWN	Responsible party for the dependency.
evidence	Yes for assurance	Supporting documents, observations or tests.
evidenceStatus	Yes	Evidence status (§53).
confidence	Yes for assurance	CONF-0 to CONF-4 with rationale, and CONFLICTING/STALE flags where set.
validity	Yes	Effective period and validation date.

38. Dependency classes

Class	Definition	Examples
Electrical	Reliance on electrical supply, quality, protection or distribution.	GPU rack -> rack PDU -> busway -> UPS path.
Thermal	Reliance on heat capture, transport or rejection.	GPU tray -> cold plate -> secondary loop -> CDU.
Hydraulic	Reliance on fluid path, pressure, flow or quality.	CDU -> facility water loop.
Mechanical/structural	Reliance on physical support, airflow or mechanical function.	Rack-scale system -> floor loading and anchoring.
Network	Reliance on connectivity, bandwidth, route or fabric service.	Compute node -> scale-out fabric.
Control-system	Reliance on sensors, automation, logic or command.	Cooling valve -> PLC -> sensor chain.
Timing/information	Reliance on time, name, address, telemetry or configuration.	Cluster service -> DNS/NTP/IPAM.

Class	Definition	Examples
Human/procedural	Reliance on competent action, approval, handover or procedure.	Generator response -> operator and escalation.
Supplier/contractual	Reliance on external service, warranty, spares or service level.	Storage recovery -> OEM support.
Geographic/environmental	Reliance on location, ambient or external condition.	Heat rejection -> outdoor wet-bulb condition.
Capacity	Reliance on sufficient headroom or reserve.	Cluster expansion -> resilient power headroom.
Recovery-only	Reliance primarily during restoration.	Compute rebuild -> golden image repository.

39. Common-mode and concentration relationships

SHARES_DEPENDENCY_WITH Two entities rely on the same material provider.

SHARES_FAULT_DOMAIN_WITH Two entities are exposed to a declared common fault domain (formerly IN_SAME_FAULT_DOMAIN_AS).

DIVERSE_FROM Two implementations differ in a declared material characteristic.

INDEPENDENT_OF An evidence-backed claim of relevant independence under stated conditions.

REDUNDANT_WITH An alternative intended to preserve a function.

CONCENTRATES An entity aggregates capability, capacity or dependency consequence.

SINGLE_POINT_FOR Failure of an entity defeats a required outcome without effective alternative.

ISOLATED_FROM A tested or designed separation relevant to a declared hazard or threat.

Part IX - Capacity, redundancy, fault-domain and resilience semantics

40. Capacity model

Capacity term	Canonical meaning
Nameplate capacity	Manufacturer or design rating before contextual adjustment.
Installed capacity	Capacity physically present.
Commissioned capacity	Capacity demonstrated within commissioning scope.
Available capacity	Capacity currently usable after condition, derating and outage.
Committed capacity	Capacity allocated, contracted or reserved.
Demand	Observed or forecast requirement under stated condition.
Headroom	Available capacity less demand and commitments, subject to constraints.
Resilient capacity	Capacity available during a defined adverse or maintenance condition.
Stranded capacity	Installed capacity unusable because another dependency constrains it.
Transient capacity	Capacity supportable for a limited duration.
Recovery capacity	Capacity available while restoration is underway.
UNKNOWN capacity	Capacity cannot be established with sufficient evidence.

41. Capacity assertion schema

IOT-REQ-022 A capacity value shall identify metric, unit, entity, operating state, environmental conditions, duration, derating assumptions, evidence date and confidence.

IOT-REQ-023 Nameplate, installed, commissioned, available, committed, resilient and uncommitted capacity shall not be used interchangeably.

IOT-REQ-024 A cross-domain capacity claim shall identify the limiting dependency and any stranded capacity.

42. Redundancy semantics

Concept	Required elements
Redundancy claim	Protected function, alternatives, required capacity, failure condition, transfer mechanism, transfer time and evidence.
Diversity claim	Characteristic in which alternatives materially differ, such as route, technology, supplier or control.
Independence claim	Relevant common causes considered, shared dependencies identified, applicable states and evidence.
Maintainability claim	Function sustained while defined assets are removed from service safely.
Concurrent-maintenance claim	Specific maintenance scenario, capacity, operational restrictions and evidence.
Fault tolerance claim	Failure set, containment boundary, residual outcome and evidence.
Graceful degradation	Priority, reduced outcome, trigger, control mechanism and recovery path.

43. Fault-domain taxonomy

- Physical: room, rack, route, fire zone, flood zone or structural area.
- Electrical: utility source, transformer, switchboard, UPS path, busway or protection zone.
- Thermal: air zone, coolant loop, CDU, chiller train, heat-rejection path.
- Network: switch, plane, fabric, route, controller or failure group.
- Storage: node group, enclosure, metadata service, protection set or site.
- Compute: node, tray, rack, scale-up domain, cluster partition or scheduler domain.
- Control-system: sensor chain, controller, automation logic, communication path or management platform.
- Software/configuration: firmware baseline, driver, image, policy or configuration source.
- Human/operational: team, shift, procedure, supplier or authority concentration.
- External: utility, carrier, water source, fuel supply, vendor or geographic region.

44. Resilience claim record

Property	Meaning
subject	Capability, outcome, function or system.
adverseCondition	Failure, threat, hazard, maintenance or compound scenario.
initialState	State before the event.
requiredOutcome	Minimum acceptable service result.
permittedDegradation	Allowed reduction and duration.
safeguards	Measures expected to act.
responseSequence	Detection, decision, transfer, containment and recovery sequence.
recoveryObjective	Target restoration condition.
evidence	Tests, events, analysis or observations.
limitations	Uncovered states, assumptions and constraints.
confidence	Supported conclusion and rationale.

Part X - Operating state, temporal state and change semantics

45. Operating state taxonomy

Code	Definition
NORMAL	Approved architecture operates within declared envelope; no material planned impairment.
DEGRADED	One or more functions, safeguards, paths or reserves are impaired while some capability remains.
MAINTENANCE	A planned intervention changes availability, configuration, staffing or safeguards.
EMERGENCY	Exceptional condition requiring urgent protective action.
RECOVERY	Controlled restoration and validation after disruption.
TRANSITION	Movement between architectures, states, locations or control regimes.
SHUTDOWN	Intentional cessation with controlled residual dependencies.
FAILED	Required function is not delivered.
UNKNOWN	Current operating state cannot be determined.

46. State transition model

> **IOT-REQ-025** State transitions shall identify trigger, authority, preconditions, required safeguards, expected duration, monitoring, exit criteria and evidence.
>
> **IOT-REQ-026** Temporary bypass and maintenance configurations shall be represented as explicit states or configuration versions.
>
> **IOT-REQ-027** Assurance claims shall identify the operating states they cover and shall not be generalised to uncovered states.

47. Temporal truth

Temporal construct	Use
Observation time	When a fact was observed.
Effective time	When a fact or relationship applies in the represented system.
Record time	When the information entered the repository.

Temporal construct	Use
Validity interval	Start and end of applicability.
Evidence period	Period covered by evidence.
Review date	When a record was last assessed.
Expiry date	When evidence or an assertion is no longer presumed current.
Supersedes	Links a new record or baseline to the prior one.
Derived from	Links a conclusion to source records.

48. Configuration perspectives

Perspective	Meaning
Intended	Approved requirement or design intent.
Designed	Documented architecture.
Procured	Contracted or ordered implementation.
Installed	Physical or logical implementation present.
As-built	Recorded installed configuration.
Commissioned	Configuration subjected to defined acceptance tests.
Observed	State directly observed or measured.
Current approved	Authoritative operational baseline.
Temporary	Time-bounded deviation or workaround.
Future	Planned target state.
Retired	No longer active but retained for history and dependency closure.

49. Change Event schema

Field	Description
changeId	Stable identifier.
changeType	Addition, removal, replacement, upgrade, reconfiguration, relocation, capacity change, supplier change or emergency change.
affectedEntities	Assets, systems, interfaces, outcomes and dependencies.
preState/postState	Configuration and operating state before and after.
impactAnalysis	Affected capacity, fault domains, safeguards, evidence and claims.
authority	Approver and implementer.
plannedWindow	Applicable schedule if planned.
validation	Required testing and acceptance.
rollback	Restoration path and conditions.
assuranceImpact	Claims invalidated, limited or requiring reassessment.

Part XI - Responsibility, evidence, confidence and UNKNOWN semantics

50. Responsibility model

Relationship	Meaning	AIDCF-014 exchange form
OWNS	Legal, financial or accountable ownership.	GOVERNED_BY (inverse) with role = owner
OPERATES	Performs routine operation.	OPERATED_BY (inverse)
CONTROLS	Holds authority to command or configure.	CONTROLLED_BY (inverse)
MAINTAINS	Performs inspection, servicing and repair.	OPERATED_BY (inverse) with role = maintainer
SUPPLIES	Provides a resource, product or service.	DEPENDS_ON or CONSUMES_RESOURCE from the consumer
MONITORS	Observes state and raises information or alerts.	MONITORED_BY (inverse)
APPROVES	Authorises design, change, work or decision.	APPROVED_BY (inverse)
ASSURES	Performs independent or internal assurance activity.	ASSESSED_BY or REVIEWED_BY (inverse)
ACCEPTS_RISK_FOR	Holds authority to accept residual exposure.	APPROVED_BY (inverse) on the risk-acceptance record
CUSTODIAN_OF	Maintains an evidence or information object.	Custodian attribute of EvidenceObject
RESPONDS_TO	Has response responsibility for an event or alarm.	OPERATED_BY (inverse) with role = responder
ESCALATES_TO	Transfers information or decision authority.	GOVERNED_BY with role = escalation

51. Shared-responsibility record

> **IOT-REQ-028** A shared-responsibility allocation shall identify function, assets, lifecycle stages, operating states, responsible party, accountable party, consulted parties, evidence custodian and unresolved gaps.
>
> **IOT-REQ-029** Contractual responsibility shall not be treated as proof of operational capability or control effectiveness.

52. Evidence attachment model

Evidence relation	Meaning	AIDCF-014 exchange form
SUPPORTS_ASSERTION	Evidence supports a testable statement.	SUPPORTS
CHALLENGES_ASSERTION	Evidence contradicts or weakens a statement.	CHALLENGES
QUALIFIES_ASSERTION	Evidence limits the scope or conditions of a statement.	QUALIFIES
OBSERVES_ENTITY	Evidence records the state of an entity.	EVIDENCED_BY (inverse, from the entity)
VALIDATES_SAFEGUARD	Evidence demonstrates a safeguard under stated conditions.	SUPPORTS on the assertion about the safeguard
ESTABLISHES_PROVENANCE	Evidence identifies source, origin or custody.	Provenance attributes (AIDCF-014 §26)
COVERS_PERIOD	Evidence applies to a declared period.	period attribute
COVERS_STATE	Evidence applies to one or more operating states.	state attribute
CONFLICTS_WITH	Evidence is inconsistent with another evidence object.	CHALLENGES between assertions, with CONFLICTING flag
SUPERSEDES	Newer controlled evidence replaces an earlier record.	SUPERSEDES

53. Evidence status and confidence

Two distinct properties are recorded. **Evidence status** describes how a record or assertion is currently supported; it is a property of the information. **Confidence** is the assessor's bounded belief in an assurance claim and uses the canonical AIDCF-008 scale; it is a property of a conclusion. Evidence status informs confidence but does not determine it, and neither is a control result.

Evidence status	Meaning
CONFIRMED	Direct, relevant and sufficiently current evidence supports the assertion.
PARTIALLY_CONFIRMED	Evidence supports part of the assertion or scope.
ASSERTED	Claim provided by an accountable source but not independently corroborated.
INFERRED	Derived from related facts using declared reasoning.
CONFLICTING	Material sources disagree and conflict remains unresolved.
STALE	Evidence or observation is no longer sufficiently current.
UNKNOWN	Information is insufficient to reach a defensible state.

Confidence	Meaning (AIDCF-008)
CONF-0 UNKNOWN	No defensible confidence state.
CONF-1 LOW	Weak or limited evidence; high sensitivity to assumptions.
CONF-2 MODERATE	Reasonable support with material limitations.
CONF-3 HIGH	Strong support with no unresolved material conflict within scope.
CONF-4 VERY_HIGH	Highly defensible, independently corroborated and representative support for the bounded claim.

The flags CONFLICTING and STALE may be attached to any confidence value. v1.0 used the evidence-status codes above as "confidence states"; records migrating from v1.0 move them to evidenceStatus and assign confidence separately.

54. UNKNOWN doctrine

Materiality is defined in AIDCF-002 §22 and used here unchanged: a fact, dependency, condition, deficiency or UNKNOWN is **MATERIAL** if, had it been known or resolved differently, a competent assessor would reasonably expect it to change (a) a control result, (b) a gate result, (c) the decision state, or (d) a condition, limitation or ceiling attached to the decision, for the declared Capability Instance, Service Outcomes and operating states. Wherever this standard says "material", that test applies.

> **IOT-REQ-030** UNKNOWN may apply to an entity existence, class, property, relationship, state, capacity, owner, evidence status or conclusion.
>
> **IOT-REQ-031** UNKNOWN shall be recorded as an UnknownRecord (Part XV) identifying the missing information, why it is MATERIAL or not, current owner for resolution where known, UNKNOWN type (U1–U10), severity (US1–US5) and assurance impact.
>
> **IOT-REQ-032** UNKNOWN shall not be replaced with an assumed favourable value to enable scoring, a gate PASS or a favourable decision state.

55. Conflict representation

Conflicting records shall be retained with separate provenance and effective periods. A resolution record may select, reconcile or reject assertions, but the original conflict trail should remain auditable.

Part XII - Conformance rules and validation shapes

56. Core validation rules

Rule	Validation statement
IOT-VAL-001	Every Capability Instance has at least one Service Outcome.
IOT-VAL-002	Every Service Outcome REQUIRES_FUNCTION at least one Required Function.
IOT-VAL-003	Every Required Function is IMPLEMENTED_BY at least one System, Asset or external service, or has an UnknownRecord.
IOT-VAL-004	Every material physical Asset has a Location or UNKNOWN.
IOT-VAL-005	Every material Asset has an owner and operator or explicit UNKNOWN.
IOT-VAL-006	Every boundary-crossing dependency identifies an external provider or UNKNOWN.
IOT-VAL-007	Every capacity assertion has metric, unit, state, condition, time and confidence.
IOT-VAL-008	Every redundancy claim identifies protected function, alternatives, transfer mechanism, common dependencies and evidence.
IOT-VAL-009	Every independence claim is evidence-backed and condition-bounded.
IOT-VAL-010	Every critical dependency identifies applicable states and strength.
IOT-VAL-011	Every assurance-relevant relationship has provenance and effective period.
IOT-VAL-012	Every asset class derives from a canonical or registered extension class.
IOT-VAL-013	Every Change Event links affected entities and assurance impact.
IOT-VAL-014	Every UNKNOWN record states the unresolved attribute and its consequence.
IOT-VAL-015	Every Service Outcome and resilience claim is tied to a Capability Instance and Operating Envelope.
IOT-VAL-016	No evidence object is treated as a safeguard; it evidences a safeguard or assertion.
IOT-VAL-017	No control abstraction is represented as proof that a real safeguard exists.
IOT-VAL-018	No maturity value automatically creates an assurance decision.
IOT-VAL-019	No excluded dependency is removed from the capability dependency view.
IOT-VAL-020	No vendor-specific term replaces its canonical parent in cross-profile exchange.
IOT-VAL-021	Every class instance carries a valid primary domain code or the Upper tag, consistent with Appendix A.
IOT-VAL-022	Every ControlAssessment has exactly one result, one claim type, a scope, operating states and an evidence package or UnknownRecord.
IOT-VAL-023	Every GateResult status is one of PASS, FAIL, UNKNOWN, APPROVED_NA or NOT_ASSESSED.
IOT-VAL-024	Every AssuranceDecision has a decision state, lifecycle state, authority, scope, validity and links to its GateResults and UnknownRecords.
IOT-VAL-025	Every IACS Zone and Conduit in scope declares SL-T; any claimed SL-A is evidence-backed and never copied from SL-C.

57. Quality tests

Test family	Questions
Completeness	Are material capabilities, outcomes, functions, assets, external services and dependencies represented?
Consistency	Do class and relationship uses obey definitions and cardinalities?
Traceability	Can each material assertion be traced to evidence and provenance?
Currency	Do effective dates and configuration states reflect the assessment point in time?
Boundary integrity	Are exclusions and boundary-crossing dependencies visible?
Cross-domain integration	Do power, cooling, control, compute, storage and network chains connect?
Failure readiness	Can AIDCF-004 attach causes, failure modes, propagation and consequences without semantic gaps?
Assurance readiness	Can controls, evidence, results, findings, UNKNOWN Records, gate results and decisions attach to the represented subject?
Interoperability	Can the dataset be exported using controlled identifiers and namespaces?
Human usability	Can engineers and assessors interpret the model without specialist graph tooling?

58. Prohibited modelling practices

- Using a single building-level rating as a property of every hosted capability.
- Treating product type as proof of performance or compliance.
- Hiding external dependencies because they are contractually assigned.
- Representing nominal redundancy without shared-dependency analysis.
- Combining designed, as-built and observed states into one unqualified record.
- Encoding UNKNOWN as zero, false, PASS, NONCONFORMING or NOT_APPLICABLE.
- Creating vendor-specific top-level classes where a vendor-neutral parent is sufficient.
- Using transitive graph reasoning to infer capacity, safety or engineering adequacy.
- Removing superseded records needed to understand historical decisions.

- Treating an ontology completeness score as infrastructure assurance.

Part XIII - Reference implementation profiles

59. Profile construction method

A profile selects mandatory classes, relationships, properties, evidence expectations and validation rules for a repeatable deployment pattern. Profiles shall not alter upper ontology meaning.

Profile	Scope and emphasis
Enterprise inference facility	Low- or moderate-scale inference in an enterprise site. Emphasis: latency, service networks, shared power/cooling, tenant interaction, recoverability.
Distributed training cluster	Multi-node accelerated training. Emphasis: scale-up/scale-out fabrics, checkpoint storage, job completion, power/thermal concentration and failure containment.
Liquid-cooled rack-scale AI system	High-density rack-scale system. Emphasis: direct-liquid loops, CDU dependencies, water quality, leak boundaries, transient response, rack power and structural load.
AI colocation tenancy	Tenant equipment hosted in provider facility. Emphasis: boundary, responsibility, evidence access, contracted capacity, shared fault domains and cross-connects.
Sovereign AI infrastructure	Strategic capability with location and control constraints. Emphasis: operational autonomy, remote support, jurisdiction, supply, spares, identities and external services.
AI supercomputing facility	Large HPC/AI deployment. Emphasis: high-performance fabric, parallel storage, scheduler, job resilience, large-scale power/cooling and recovery.
Edge AI facility	Constrained remote deployment. Emphasis: environmental envelope, limited staffing, network dependence, remote management and local safety.
Multi-site AI service	Capability distributed across sites. Emphasis: inter-site dependencies, data replication, routing, control plane, failover and service-level boundary.

60. Worked profile: liquid-cooled distributed training capability

Element	Illustrative ontology record
Capability Instance	Distributed foundation-model training capability at Site A.
Service Outcomes	Minimum usable accelerator capacity; maximum interruption; checkpoint survival; bounded restart time.
Required Functions	Electrical continuity; heat capture and rejection; scale-out transport; metadata access; checkpoint persistence; scheduling.
Systems	Compute cluster; rack power path; liquid cooling system; parallel storage; AI fabric; management system.
Boundary	Defined halls, racks, infrastructure systems and external utility/OEM services.
Cross-domain dependencies	Utility -> UPS -> busway -> rack; cold plate -> secondary loop -> CDU -> facility loop -> heat rejection; node -> fabric -> storage.
Fault domains	Rack-scale compute domain; CDU loop; electrical path; fabric plane; storage protection group; operator team.
Operating states	NORMAL; MAINTENANCE (single electrical path); DEGRADED (cooling); RECOVERY; TRANSITION.
Material UNKNOWN examples	Unverified secondary-loop autonomy; unclear shared controller; stale resilient-capacity study (each an UnknownRecord with type and severity).
Evidence links	As-built diagrams, commissioning tests, telemetry, maintenance records, failure tests and observed configuration.

61. Competency questions for profile review

- Can every required service outcome be traced to functions, systems and assets?
- Are physical and logical topology views consistent?
- Are power and cooling capacities expressed under the same operating conditions as compute demand?
- Are management and control dependencies represented, including those required during recovery?
- Are shared dependencies visible across apparent redundant paths?
- Does evidence cover degraded and maintenance states, not only normal operation?
- Can an external reviewer identify what remains UNKNOWN without reverse-engineering narrative reports?

Part XIV - Machine-readable publication specification

62. Required publication package

File	Purpose
aidcf-iot-v1.0.owl	Formal ontology in OWL.
aidcf-iot-v1.0.ttl	Human-readable RDF/Turtle serialization.
aidcf-context-v1.0.jsonld	JSON-LD context.
aidcf-schema-v1.0.json	JSON Schema for exchange records.
aidcf-shapes-v1.0.ttl	SHACL validation shapes.
aidcf-vocabulary-v1.0.csv	Controlled term register, including domain codes and Appendix A tags.
aidcf-relationships-v1.0.csv	Relationship registry.
aidcf-example-minimal-v1.0.jsonld	Minimal conformant example.

File	Purpose
aidcf-example-training-v1.0.jsonld	Worked training-cluster profile.
CHANGELOG.md	Version history and changes.
MIGRATION.md	Compatibility and migration guidance.
LICENSE.md	Usage and extension rules.

63. Namespace policy

The AIDCF ontology namespace is `https://w3id.org/aidcf/ontology#` with the prefix `aidcf:`, and the version IRI of the current release line is `https://w3id.org/aidcf/ontology/1.3`. Registration of the `aidcf` space with the w3id.org persistent-identifier service is pending; until it is confirmed these IRIs shall be treated as reserved draft identifiers and shall not be represented as resolvable production identifiers. v1.0 proposed `https://w3id.org/aidcf/iot/1.0/`; that pattern is withdrawn. The class and relationship names in this standard are the local names under this namespace (for example `aidcf:CapabilityInstance`).

64. Identifier rules

- Class identifiers use PascalCase, for example CapabilityInstance.
- Relationship identifiers use UPPER_SNAKE_CASE in human registries and stable lowerCamelCase or IRI form in machine schemas.
- Instance identifiers shall be opaque enough to avoid embedding sensitive facility details.
- Identifiers shall not be reassigned after retirement.
- Human-readable labels may change without changing semantic identifiers where meaning is stable.
- Breaking semantic changes require a new major namespace version.

65. JSON exchange skeleton

Illustrative only; the context IRI resolves only once w3id registration is complete (§63).

{
"@context": "https://w3id.org/aidcf/ontology/1.3/context.jsonld",
"id": "capability-instance:example-001",
"type": "CapabilityInstance",
"name": "Example distributed training capability",
"serviceOutcomes": ["outcome:compute-capacity", "outcome:checkpoint-recovery"],
"assessmentBoundary": "boundary:example-001",
"operatingEnvelope": "envelope:example-001",
"requiredFunctions": ["function:power-continuity", "function:heat-removal", "function:scale-out-transport"], "evidenceStatus": "ASSERTED",
"confidence": "CONF-1"
}

66. Extension governance

> **IOT-REQ-033** An extension shall publish namespace, owner, scope, parent mappings, definitions, constraints, examples, tests, version and compatibility statement.
>
> **IOT-REQ-034** A profile shall not weaken canonical validation rules without declaring a nonconformance.
>
> **IOT-REQ-035** Vendor mappings should map product terms to canonical classes rather than creating competing upper ontologies.

67. Integration boundary with AI Trust Graph

AIDCF-005 specifies the integration in full; the summary below fixes only the ontology boundary.

AIDCF primary semantics	Shared handoff	AI Trust Graph primary semantics
Site, building, power, cooling, facility OT, physical assets, compute hardware, storage infrastructure and interconnect.	Identifiers, evidence, infrastructure services, machine identities, management interfaces, dependencies and findings.	Models, agents, tools, APIs, human and machine identities, authority, reachability and digital trust paths.

Part XV - Assurance-chain classes

Parts III to XI represent the infrastructure half of the canonical chain (steps 2 to 6) and the attachment points for adverse scenarios and safeguards (steps 7 and 8). This Part represents the assurance half: the decision question (step 1), controls and evidence (steps 9 to 11), results, findings and UNKNOWN Records (steps 12 and 13), gates, scores and decisions (steps 14 to 16), and the report and reassessment trigger that close the loop. Meanings are those of AIDCF-002; controlled values are owned by the artifacts named in §69. Class names are identical to the AIDCF-014 §7 object classes except where §68 states otherwise.

68. Class definitions

Class	Definition	Key attributes	AIDCF-014 class
aidcf:DecisionQuestion	The precise question an assessment exists to answer, fixing subject, intended use, decision owner and date of reliance; the context of the assurance-case top claim.	questionText; capabilityInstanceRef; intendedUse; decisionOwner; asOfDate; serviceOutcomeRefs; operatingStates	Carried as the decisionQuestion attribute of Assessment and AssuranceDecision
aidcf:ControlDomain	One of the fifteen top-level groupings of the AIDCF-007A registry, identified by a fixed three-letter code (GOV … EVA).	domainId; title; familyIds (exactly six); registryVersion	ControlDomain
aidcf:ControlFamily	A group of four related controls within one domain, identified as `<DOM>-01` to `<DOM>-06`.	familyId; domainId; title; controlIds (exactly four)	ControlFamily
aidcf:Control	An assessable requirement in the AIDCF-007A registry, identified as `MCF-<DOM>-NNN`, stating objective, requirement, assessment criteria and evidence profile. A control is evaluated against one or more Safeguards; it is not itself a protective measure.	controlId; familyId; objective; requirement; assessmentCriteria; parameters; evidenceProfile (claimType, profileTier, sufficiencyFloor); assessmentMethods; controlType; requiredAssuranceLevel; gateMappings (gateId, role CRITICAL or SUPPORTING); ontologyClasses; applicability	Control
aidcf:EvidenceRequirement	The evidence a control needs for a stated claim type: required and optional evidence classes, quality floor, sufficiency floor, state and period coverage, independence and mandatory items.	evidenceRequirementId; controlRef; claimType; requiredClasses (E1–E7); qualityFloor (P F C R T I); sufficiencyFloor (S2–S5); stateCoverage; periodCoverage; independence; mandatoryItems; acceptableAlternatives; unknownEffect	EvidenceRequirement
aidcf:EvidenceObject	A record, observation, measurement, test output or analysis used to support, challenge or qualify one or more Assertions.	evidenceId; evidenceClass (E1–E7); source; collector; custodian; observationTime; period; operatingStates; scope; integrity (I0–I5); quality vector (P, F, C, R, T, I each 0–5); qualityBand (Q0–Q5); assertionRefs; restrictions	EvidenceObject
aidcf:EvidencePackage	A versioned, integrity-protected set of Evidence Objects assembled for a defined control, gate or decision, with a manifest and a sufficiency rating.	packageId; purposeRef (control, gate or decision); members; manifestHash; sufficiency (S0–S5); coverage (scope, states, period); gaps; reviewer	EvidencePackage
aidcf:Assertion	One precise, testable proposition, expressed as subject, predicate and object with scope, conditions and effective time, that evidence supports, challenges or qualifies.	assertionId; subjectRef; predicate; objectRef or value; scope; conditions; effectivePeriod; status (CANDIDATE, SUPPORTED, CHALLENGED, QUALIFIED, CONFLICTING, UNKNOWN, SUPERSEDED); evidenceRefs; confidence	No standalone class: carried through assertionRefs on EvidenceObject and the claim on ControlAssessment, with status from the AIDCF-014 Assertion vocabulary
aidcf:ControlAssessment	The evaluation of one Control for one claim type over a declared scope, set of operating states and period, yielding exactly one control result.	controlAssessmentId; controlRef; claimType; scope; operatingStates; period; methods; evidencePackageRef; result; confidence (CONF-0–CONF-4); findingRefs; unknownRefs; assessor; reviewer; lifecycleState (including SUSPENDED)	ControlAssessment
aidcf:Finding	A recorded condition, against a stated criterion, that is a deficiency or a MATERIAL observation, with its cause, consequence and effect on gates and scores.	findingId; criterionRef; condition; evidenceRefs; cause; consequence; classification; gateEffect; scoreEffect; owner; status; closureRef	Finding
aidcf:UnknownRecord	A preserved record that a MATERIAL attribute, relationship, result or criterion cannot be determined, with its type, severity and decision effect.	unknownId; subjectRef; missingAttribute; reason; unknownType (U1–U10); severity (US1–US5); decisionEffect; evidenceNeeded; owner; trigger; status	UnknownRecord
aidcf:CriticalGate	One of the fifteen non-compensable conditions CG-01 to CG-15, with its protected scope, criteria and evidence floor.	gateId; name; scope; criteria (AIDCF-010); evidenceFloor (S3–S5); criticalControlRefs (from AIDCF-007A gate mappings); decisionCeilings	CriticalGate
aidcf:GateResult	The evaluation of one CriticalGate for one assessment, with a status of PASS, FAIL, UNKNOWN, APPROVED_NA or NOT_ASSESSED. There is no conditional gate status.	gateResultId; gateRef; assessmentRef; applicability; criterionResults; criticalControlResults; supportingControlEvaluations; evidenceRefs; stateCoverage; findingRefs; unknownRefs; status; decisionEffect; reviewer	GateResult
aidcf:ScoreSnapshot	An immutable record of scores computed under AIDCF-010 at a point in time: control state values, family and domain means, coverage and bands. Scores never override gates.	snapshotId; assessmentRef; profileVersion; inputs (ControlAssessment refs); weights; familyScores; domainScores; coverage; band (B1–B5 or withheld); withheldReason; computedAt	ScoreSnapshot
aidcf:DecisionRecommendation	The assessor's proposed decision state, with conditions, limitations and ceilings derived from Gate Results, UNKNOWN severities and scores, submitted for review and approval.	recommendationId; assessmentRef; proposedState; gateCeilings; unknownCeilings; conditions; limitations; rationale; recommender	DecisionRecommendation
aidcf:AssuranceDecision	The accountable, authorised conclusion answering the Decision Question, in one decision state, with its authority, scope, conditions and validity.	decisionId; decisionQuestion; subjectRef; scope; recommendationRef; gateResultRefs; unknownRefs; scoreSnapshotRef; conditions; decisionState; lifecycleState; authority; validity	AssuranceDecision
aidcf:AssuranceStatement	The controlled wording that expresses an approved Assurance Decision for a named audience, with mandatory bounds and limitations (AIDCF-012).	statementId; decisionRef; audience; wording; bounds; limitations; issuer; issueDate	AssuranceStatement
aidcf:AssuranceReport	The issued report carrying one or more Assurance Statements and their basis: results, findings, UNKNOWN register, gate results, scores and limitations (AIDCF-012).	reportId; reportClass; programRef; assessmentRef; decisionRefs; statementRefs; intendedUsers; classification; validity; approvals; distributionControl	AssuranceReport
aidcf:ReassessmentTrigger	A condition or event that invalidates, limits or requires review of prior results or a decision, such as a material change, incident, evidence expiry, UNKNOWN resolution or monitoring threshold.	triggerId; triggerType; rule or threshold; subjectRefs; affectedObjectRefs; requiredAction; raisedAt; status	ReassessmentTrigger

69. Controlled values used by assurance classes

The controlled values below are owned by the artifacts named and are listed here so that the ontology can type them. This standard does not redefine them.

Attribute	Controlled values	Owner
claimType	DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS, RESILIENCE	AIDCF-002; AIDCF-007A
requiredAssuranceLevel	AL2 to AL5 (programme levels AL0 to AL6)	AIDCF-011
result (ControlAssessment)	CONFORMING, PARTIAL, NONCONFORMING, UNKNOWN, NOT_APPLICABLE, NOT_ASSESSED	AIDCF-002; AIDCF-010
status (GateResult)	PASS, FAIL, UNKNOWN, APPROVED_NA, NOT_ASSESSED	AIDCF-010
decisionState	REASONABLY_ASSURED, CONDITIONALLY_ASSURED, LIMITED, NOT_ASSURED, INCONCLUSIVE	AIDCF-010
lifecycleState (AssuranceDecision)	DRAFT, RECOMMENDED, APPROVED, ACTIVE, SUSPENDED, WITHDRAWN, SUPERSEDED, EXPIRED	AIDCF-010
confidence	CONF-0 UNKNOWN to CONF-4 VERY_HIGH, with flags CONFLICTING and STALE	AIDCF-008
evidenceClass; quality; qualityBand; sufficiency; integrity	E1–E7; P, F, C, R, T, I rated 0–5; Q0–Q5; S0–S5; I0–I5	AIDCF-008
unknownType; severity	U1–U10; US1 Low to US5 Systemic	AIDCF-008; AIDCF-002 §31
methods	INQUIRY, DOCUMENT_REVIEW, INSPECTION, OBSERVATION, RECONCILIATION, ANALYSIS, SAMPLING, TEST, REPERFORMANCE, SCENARIO_EXERCISE	AIDCF-009
controlType	DIRECTIVE, PREVENTIVE, DETECTIVE, CORRECTIVE, RECOVERY	AIDCF-007A
band	B5 Strong to B1 Very weak, or "Band withheld (coverage/UNKNOWN)"	AIDCF-010

70. Assurance-chain relationships

All predicates below are AIDCF-014 §9 relationship types.

Subject	Predicate	Object	Meaning
Safeguard	GOVERNED_BY	Control	The safeguard is the real measure that the control evaluates.
ControlDomain	COMPOSED_OF	ControlFamily	A domain holds six families.
ControlFamily	COMPOSED_OF	Control	A family holds four controls.
Control	HAS_EVIDENCE_REQUIREMENT	EvidenceRequirement	The control's claim needs this evidence.
EvidenceRequirement	EVIDENCED_BY	EvidencePackage	The package is offered against the requirement.
EvidencePackage	COMPOSED_OF	EvidenceObject	Package membership.
EvidenceObject	SUPPORTS / CHALLENGES / QUALIFIES	Assertion	Evidence bears on a precise proposition.
Control	ASSESSED_BY	ControlAssessment	The control was evaluated for a claim type and scope.
ControlAssessment	HAS_FINDING / HAS_UNKNOWN	Finding / UnknownRecord	Deficiencies and indeterminacies arising from the evaluation.
GateResult	EVALUATES_GATE	CriticalGate	The result is the evaluation of that gate.
ControlAssessment	SUPPORTS or CHALLENGES	GateResult	A CRITICAL or SUPPORTING control result informs the gate evaluation.
Finding / UnknownRecord	CONSTRAINS	GateResult / AssuranceDecision	Defeaters and ceilings on the gate or the decision.
ControlAssessment	SUMMARISED_BY	ScoreSnapshot	Scored inputs to the snapshot.
DecisionRecommendation	RECOMMENDS	AssuranceDecision	Proposed state, conditions and ceilings.
AssuranceDecision	REVIEWED_BY / APPROVED_BY	Role	Independent review and decision authority.
AssuranceDecision	REPORTED_IN	AssuranceReport	The decision is expressed in the report through an AssuranceStatement.
Finding	REQUIRES_ACTION / CLOSED_BY	Action	Remediation and closure.
ReassessmentTrigger	TRIGGERS_REASSESSMENT	ControlAssessment / GateResult / AssuranceDecision	The trigger requires review of the object.
AssuranceDecision	INVALIDATED_BY / SUPERSEDES	ReassessmentTrigger / AssuranceDecision	Loss of validity and replacement without erasure.

71. Requirements for assurance-chain records

> **IOT-REQ-036** Implementations shall represent controls, evidence, results, findings, UNKNOWN Records, gate results, scores, recommendations, decisions, statements, reports and reassessment triggers using the classes of §68 and the AIDCF-014 class names they map to.
>
> **IOT-REQ-037** A ControlAssessment shall record exactly one result for exactly one claim type, scope, set of operating states and period; a control assessed for several claim types shall have one ControlAssessment per claim type.
>
> **IOT-REQ-038** A ControlAssessment whose evidence is below the floor of its EvidenceRequirement shall carry the result UNKNOWN and link an UnknownRecord; it shall not carry a reduced numeric value in place of the result.
>
> **IOT-REQ-039** A GateResult shall carry one of PASS, FAIL, UNKNOWN, APPROVED_NA or NOT_ASSESSED; conditions shall be represented only on the DecisionRecommendation and AssuranceDecision.

IOT-REQ-040 An AssuranceDecision shall reference the DecisionRecommendation, GateResults, UnknownRecords and ScoreSnapshot on which it relies, and a ScoreSnapshot shall not be the sole basis of a decision state.

IOT-REQ-041 An Evidence Object shall not be typed as a Safeguard or a Control; it evidences an Assertion about a Safeguard, an asset or a result.

Part XVI - Industrial automation and control system security concepts (IEC 62443 alignment)

AI data centres depend on industrial automation and control systems for their Required Functions: the BMS, EPMS and SCADA platforms, programmable logic controllers in coolant distribution units, chiller plants and switchgear, generator and transfer-scheme controllers, fire-alarm interfaces, and the engineering workstations and remote-support paths used to change them. The IEC 62443 series provides the established vocabulary for securing such systems. This Part imports the concepts needed to represent them; it paraphrases and references the series at document level and does not reproduce its requirements. Conformity with IEC 62443 is not claimed or implied by using these classes.

72. IACS classes

Class	Definition	Key attributes	Relationships
aidcf:IACS	Industrial Automation and Control System: the personnel, hardware, software and policies that operate, monitor and protect a physical process. In AIDCF, the control systems of power, cooling, fire and building services, including their engineering workstations, historians and gateways. Subclass of InfrastructureSystem.	iacsId; controlledFunctions (RequiredFunction refs); components (BMS, EPMS, SCADA, PLC, RTU, controllers); assetOwner; integrator; maintainer; zoneRefs	CONTROLLED_BY (from POW, THM and PHY systems to the IACS); COMPOSED_OF (zones, components)
aidcf:IACSZone	A grouping of logical or physical assets that share common security requirements, defined through a security risk assessment (IEC 62443-3-2). Distinct from the physical aidcf:SecurityZone.	zoneId; members; controlledFunctions; riskAssessmentRef; slTarget; slAchieved; owner; boundaryDescription	COMPOSED_OF (assets); PROTECTED_BY (safeguards); SHARES_FAULT_DOMAIN_WITH (where zones share platforms, identities or networks)
aidcf:Conduit	A logical grouping of communication channels connecting two or more zones that share common security requirements (IEC 62443-3-2).	conduitId; connectedZones; channels (protocols, endpoints, direction); slTarget; slAchieved; countermeasures; remoteAccess (yes/no); owner	CONNECTS_TO (zones); PROTECTED_BY (filtering, gateways, remote-access controls)
aidcf:SecurityLevel	A level 0 to 4 expressing the protection a zone, conduit or component provides against increasingly capable violation, recorded per foundational requirement as a seven-element vector. Three kinds are distinguished: **SL-T** (target) is the level required for a zone or conduit by the risk assessment; **SL-C** (capability) is the level a component or system can provide when properly configured (IEC 62443-3-3, IEC 62443-4-2); **SL-A** (achieved) is the level actually provided by the deployed zone or conduit as assessed.	kind (SL-T, SL-C, SL-A); subjectRef; vector (FR1–FR7, each 0–4); basis; evidenceRefs; assessedAt	Attribute vector on IACSZone, Conduit and Component; SL-A is EVIDENCED_BY assessment evidence
aidcf:FoundationalRequirement	One of the seven IEC 62443 foundational requirements against which security levels are expressed: FR1 Identification and authentication control; FR2 Use control; FR3 System integrity; FR4 Data confidentiality; FR5 Restricted data flow; FR6 Timely response to events; FR7 Resource availability.	frId (FR1–FR7); label	Index of the SecurityLevel vector

73. Relationship to SEC controls and to gates CG-07 and CG-15

The concepts in §72 give SEC controls in AIDCF-007A a precise subject. Gate roles (CRITICAL or SUPPORTING) are assigned per control in the AIDCF-007A registry and are not restated here.

Concept	AIDCF-007A controls that assess it	Gate relevance
IACS inventory and controlled functions	MCF-SEC-005 BMS and EPMS Security; POW-06 and THM-06 monitoring and control families; OPS-05 configuration baseline	CG-07: the gate cannot be evaluated without knowing which IACS components control which Required Functions
Zone and conduit model with SL-T	MCF-SEC-009 Security Zoning; MCF-SEC-010 Management-Plane Segmentation; MCF-SEC-011 Filtering and Gateways	CG-07 (control-system integrity) and CG-15 (bounded digital authority)
Remote and third-party access across conduits	MCF-SEC-012 Remote Access; SUP-06 remote support controls (MCF-SUP-021 to MCF-SUP-024)	CG-15
FR1 and FR2 (identification, authentication and use control)	SEC-01 controls (MCF-SEC-001 to MCF-SEC-004)	CG-15
FR3 (system integrity)	SEC-04 controls (MCF-SEC-013 to MCF-SEC-016); SUP-04 software and firmware supply	CG-07; CG-14 (change control)
FR6 (timely response to events)	SEC-05 controls (MCF-SEC-017 to MCF-SEC-020)	CG-07; CG-15
FR7 (resource availability)	POW, THM and RES controls for the controlled functions	CG-05, CG-06, CG-10 through the controlled functions
Automated or AI-enabled commands into a zone	SEC-06 controls (MCF-SEC-021 to MCF-SEC-024)	CG-07; CG-15

74. Requirements for IACS representation

IOT-REQ-042 Where an IACS controls or protects a Required Function within the assessment boundary, the dataset shall represent the IACS, its zones and conduits, and the Required Functions each zone controls.

IOT-REQ-043 Every in-scope IACSZone and Conduit shall declare an SL-T vector with the risk-assessment reference that justifies it, or carry an UnknownRecord stating that no SL-T has been established.

IOT-REQ-044 An SL-A vector shall be supported by assessment evidence of the deployed zone or conduit; the SL-C of a component shall not be recorded as the SL-A of the zone or conduit that contains it.

IOT-REQ-045 Where SL-A is below SL-T for any foundational requirement in a zone or conduit that controls a Required Function, the difference shall be recorded as a Finding linked to the affected controls and evaluated in the GateResults for CG-07 and, where identity, privilege or remote access is involved, CG-15; where SL-A has not been evidenced, an UnknownRecord shall be raised and its severity assigned under the canonical US1–US5 scale.

IOT-REQ-046 A physical aidcf:SecurityZone shall not be used as a substitute for an aidcf:IACSZone, and network adjacency shall not be represented as a Conduit unless the communication channels it carries are identified.

Appendix A - Canonical class registry

All identifiers are local names in the `aidcf:` namespace (§63). **Domain code(s)** gives the primary AIDCF-007A domain first, then secondary codes; Upper marks a cross-domain upper-ontology class. **AIDCF-014 class** gives the exchange class under which instances are serialised; a dash means the class is an ontology-level concept carried as an attribute or relationship rather than a separate exchange object.

A.1 Upper-ontology and core classes

Canonical ID	Preferred label	Definition	Domain code(s)	AIDCF-014 class
aidcf:Capability	Capability	An intended operational ability supported by infrastructure.	CAP	—
aidcf:CapabilityInstance	Capability Instance	A specific deployment bounded by ownership, location, architecture, capacity, criticality and obligations.	CAP	CapabilityInstance
aidcf:ServiceOutcome	Service Outcome	A measurable or assessable result required from a Capability Instance.	CAP	ServiceOutcome
aidcf:RequiredFunction	Required Function	A function whose loss or degradation would materially affect a Service Outcome (v1.0: CriticalFunction).	CAP	RequiredFunction
aidcf:OperatingEnvelope	Operating Envelope	The bounded conditions under which a capability or system is intended to operate.	CAP	— (attribute set of CapabilityInstance)
aidcf:AssessmentBoundary	Assessment Boundary	The declared inclusion perimeter for direct assessment.	DEP, EVA	— (scope of Assessment)
aidcf:Dependency	Dependency	A reliance by one entity upon another entity, resource, condition or decision.	DEP	Dependency
aidcf:FaultDomain	Fault Domain	A set of entities expected to be affected by a common failure.	RES, DEP	FaultDomain
aidcf:RedundancyGroup	Redundancy Group	Assets or paths deliberately arranged to preserve a function.	RES	— (attribute of FaultDomain/Dependency)
aidcf:InfrastructureSystem	Infrastructure System	A coordinated set of assets and interfaces providing one or more functions.	Upper	InfrastructureSystem
aidcf:AssetInstance	Asset Instance	An identifiable real-world item, service, person, procedure or resource.	Upper	AssetInstance
aidcf:Component	Component	A constituent part of an asset or system represented where material.	Upper	AssetInstance (componentOf)
aidcf:Interface	Interface	A defined point or mechanism of exchange between entities.	Upper	Interface
aidcf:Connection	Connection	A realised physical, logical, hydraulic, electrical, thermal or organisational link.	Upper	CONNECTS_TO relationship
aidcf:Resource	Resource	Consumable or constrained input required for operation.	Upper	Resource
aidcf:Location	Location	A geographic, site, building, zone, room, row, rack or logical placement.	Upper	— (LOCATED_IN target)
aidcf:Safeguard	Safeguard	A real-world measure intended to anticipate, prevent, detect, contain, respond, recover or adapt.	Upper	Safeguard
aidcf:Actor	Actor	A human or machine participant with a relevant role.	Upper	—
aidcf:Organisation	Organisation	A legally or operationally recognised party.	Upper	—
aidcf:Role	Role	A named set of responsibilities and authorities.	GOV	—
aidcf:OperatingState	Operating State	A defined condition of operation (§45).	Upper	OperatingState
aidcf:ChangeEvent	Change Event	An approved, observed or unapproved modification affecting the represented system.	OPS, CAP	Event (eventType = CHANGE)

A.2 Domain taxonomy classes

Each class below is a specialisation of the upper class appropriate to its kind (InfrastructureSystem, AssetInstance, Component, Interface, Resource, Location, Role or a record class) and is defined by its preferred label within the taxonomy family named; the family descriptions in Part V give the context. Instances are exchanged under the AIDCF-014 class of their upper class (normally AssetInstance or InfrastructureSystem; resources as Resource; records as EvidenceObject). Classes marked † were added in v1.1.

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:OperatingModel	Operating Model	GOV — Governance and authority	GOV, OPS
aidcf:Delegation	Delegation	GOV — Governance and authority	GOV
aidcf:ApprovalAuthority	Approval Authority	GOV — Governance and authority	GOV
aidcf:EscalationPath	Escalation Path	GOV — Governance and authority	GOV, OPS
aidcf:RiskAcceptance	Risk Acceptance	GOV — Governance and authority	GOV

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:Policy	Policy †	GOV — Obligations and baselines	GOV
aidcf:TechnicalStandardBaseline	Technical Standard Baseline †	GOV — Obligations and baselines	GOV
aidcf:ExceptionRecord	Exception Record †	GOV — Obligations and baselines	GOV, EVA
aidcf:ContractualObligation	Contractual Obligation †	GOV — Obligations and baselines	GOV, SUP
aidcf:CapabilityRegister	Capability Register †	CAP — Capability and demand records	CAP
aidcf:DemandProfile	Demand Profile †	CAP — Capability and demand records	CAP
aidcf:LoadStepProfile	Load Step Profile †	CAP — Capability and demand records	CAP, POW, THM
aidcf:GrowthForecast	Growth Forecast †	CAP — Capability and demand records	CAP
aidcf:DesignBasis	Design Basis	CAP — Engineering baseline and records	CAP, GOV
aidcf:SingleLineDiagram	Single-Line Diagram	CAP — Engineering baseline and records	POW, CAP, EVA
aidcf:PipingAndInstrumentationDiagram	P&ID	CAP — Engineering baseline and records	THM, CAP, EVA
aidcf:AsBuiltDrawing	As-Built Drawing	CAP — Engineering baseline and records	CAP, EVA
aidcf:EngineeringStudy	Engineering Study	CAP — Engineering baseline and records	CAP, EVA
aidcf:CommissioningScript	Commissioning Script	CAP — Engineering baseline and records	CAP, EVA
aidcf:DeviationRecord	Deviation Record	CAP — Engineering baseline and records	CAP, EVA
aidcf:ExternalService	External Service †	DEP — Boundary and external service	DEP, SUP
aidcf:BoundaryCrossing	Boundary Crossing †	DEP — Boundary and external service	DEP
aidcf:ServiceInterfaceAgreement	Service Interface Agreement †	DEP — Boundary and external service	DEP, SUP
aidcf:Site	Site	PHY — Site and external perimeter	PHY
aidcf:Campus	Campus	PHY — Site and external perimeter	PHY
aidcf:PropertyBoundary	Property Boundary	PHY — Site and external perimeter	PHY
aidcf:PerimeterBarrier	Perimeter Barrier	PHY — Site and external perimeter	PHY
aidcf:VehicleBarrier	Vehicle Barrier	PHY — Site and external perimeter	PHY
aidcf:Gate	Gate	PHY — Site and external perimeter	PHY
aidcf:GuardPost	Guard Post	PHY — Site and external perimeter	PHY
aidcf:ExternalApproach	External Approach	PHY — Site and external perimeter	PHY
aidcf:StandoffZone	Standoff Zone	PHY — Site and external perimeter	PHY
aidcf:Building	Building	PHY — Building and internal zoning	PHY
aidcf:Floor	Floor	PHY — Building and internal zoning	PHY
aidcf:DataHall	Data Hall	PHY — Building and internal zoning	PHY
aidcf:Room	Room	PHY — Building and internal zoning	PHY
aidcf:Cage	Cage	PHY — Building and internal zoning	PHY
aidcf:Suite	Suite	PHY — Building and internal zoning	PHY
aidcf:Row	Row	PHY — Building and internal zoning	PHY
aidcf:Rack	Rack	PHY — Building and internal zoning	PHY, CMP
aidcf:Cabinet	Cabinet	PHY — Building and internal zoning	PHY
aidcf:SecurityZone	Security Zone	PHY — Building and internal zoning	PHY
aidcf:RestrictedArea	Restricted Area	PHY — Building and internal zoning	PHY
aidcf:LoadingArea	Loading Area	PHY — Building and internal zoning	PHY
aidcf:StagingArea	Staging Area	PHY — Building and internal zoning	PHY
aidcf:Foundation	Foundation	PHY — Civil and structural	PHY
aidcf:FloorSlab	Floor Slab	PHY — Civil and structural	PHY
aidcf:RaisedFloor	Raised Floor	PHY — Civil and structural	PHY
aidcf:Roof	Roof	PHY — Civil and structural	PHY
aidcf:Wall	Wall	PHY — Civil and structural	PHY
aidcf:Ceiling	Ceiling	PHY — Civil and structural	PHY
aidcf:RackAnchorage	Rack Anchorage	PHY — Civil and structural	PHY
aidcf:SeismicRestraint	Seismic Restraint	PHY — Civil and structural	PHY
aidcf:LoadBearingElement	Load-Bearing Element	PHY — Civil and structural	PHY
aidcf:AccessPoint	Access Point	PHY — Entry and identity control	PHY
aidcf:Turnstile	Turnstile	PHY — Entry and identity control	PHY
aidcf:Mantrap	Mantrap	PHY — Entry and identity control	PHY
aidcf:Door	Door	PHY — Entry and identity control	PHY
aidcf:Lock	Lock	PHY — Entry and identity control	PHY
aidcf:CredentialReader	Credential Reader	PHY — Entry and identity control	PHY, SEC
aidcf:BiometricReader	Biometric Reader	PHY — Entry and identity control	PHY
aidcf:VisitorSystem	Visitor System	PHY — Entry and identity control	PHY

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:KeySystem	Key System	PHY — Entry and identity control	PHY
aidcf:Camera	Camera	PHY — Detection and surveillance	PHY
aidcf:VideoManagementSystem	Video Management System	PHY — Detection and surveillance	PHY, SEC
aidcf:IntrusionSensor	Intrusion Sensor	PHY — Detection and surveillance	PHY
aidcf:DoorContact	Door Contact	PHY — Detection and surveillance	PHY
aidcf:MotionSensor	Motion Sensor	PHY — Detection and surveillance	PHY
aidcf:SecurityAlarm	Security Alarm	PHY — Detection and surveillance	PHY
aidcf:Intercom	Intercom	PHY — Detection and surveillance	PHY
aidcf:LoadingBay	Loading Bay	PHY — Asset movement and custody	PHY
aidcf:GoodsLift	Goods Lift	PHY — Asset movement and custody	PHY
aidcf:ScreeningPoint	Screening Point	PHY — Asset movement and custody	PHY
aidcf:StorageArea	Storage Area	PHY — Asset movement and custody	PHY
aidcf:Seal	Seal	PHY — Asset movement and custody	PHY
aidcf:ChainOfCustodyRecord	Chain-of-Custody Record	PHY — Asset movement and custody	PHY, SUP, EVA
aidcf:SecurityOperationsCentre	Security Operations Centre	PHY — Physical response	PHY
aidcf:GuardForce	Guard Force	PHY — Physical response	PHY
aidcf:ResponseProcedure	Response Procedure	PHY — Physical response	PHY
aidcf:DispatchChannel	Dispatch Channel	PHY — Physical response	PHY
aidcf:IncidentReconstructionRecord	Incident Reconstruction Record	PHY — Physical response	PHY, EVA
aidcf:DetectionZone	Detection Zone	PHY — Fire and life safety	PHY
aidcf:FireAlarmPanel	Fire Alarm Panel	PHY — Fire and life safety	PHY
aidcf:SuppressionZone	Suppression Zone	PHY — Fire and life safety	PHY
aidcf:SuppressionAgentSystem	Suppression Agent System †	PHY — Fire and life safety	PHY, ENV
aidcf:SprinklerSystem	Sprinkler System	PHY — Fire and life safety	PHY
aidcf:SmokeControl	Smoke Control	PHY — Fire and life safety	PHY
aidcf:EmergencyEgress	Emergency Egress	PHY — Fire and life safety	PHY
aidcf:EmergencyPowerOff	Emergency Power Off	PHY — Fire and life safety	PHY, POW
aidcf:IncomingSupply	Incoming Supply	POW — Utility and grid interface	POW, DEP
aidcf:UtilityFeed	Utility Feed	POW — Utility and grid interface	POW, DEP
aidcf:GridConnection	Grid Connection	POW — Utility and grid interface	POW, DEP
aidcf:Substation	Substation	POW — Utility and grid interface	POW
aidcf:OnSiteGeneration	On-Site Generation	POW — Generation and fuel	POW
aidcf:Generator	Generator	POW — Generation and fuel	POW
aidcf:FuelTank	Fuel Tank	POW — Generation and fuel	POW, ENV
aidcf:FuelSystem	Fuel System	POW — Generation and fuel	POW
aidcf:StartingSystem	Starting System	POW — Generation and fuel	POW
aidcf:ExhaustSystem	Exhaust System	POW — Generation and fuel	POW, ENV
aidcf:Transformer	Transformer	POW — Conversion, energy storage and distribution	POW
aidcf:Switchgear	Switchgear	POW — Conversion, energy storage and distribution	POW
aidcf:MVSwitchgear	MV Switchgear	POW — Conversion, energy storage and distribution	POW
aidcf:LVSwitchgear	LV Switchgear	POW — Conversion, energy storage and distribution	POW
aidcf:TransferScheme	Transfer Scheme	POW — Conversion, energy storage and distribution	POW
aidcf:StaticTransferSwitch	Static Transfer Switch	POW — Conversion, energy storage and distribution	POW
aidcf:UPS	UPS	POW — Conversion, energy storage and distribution	POW
aidcf:Battery	Battery	POW — Conversion, energy storage and distribution	POW, ENV
aidcf:EnergyStorage	Energy Storage	POW — Conversion, energy storage and distribution	POW
aidcf:Busway	Busway	POW — Conversion, energy storage and distribution	POW
aidcf:PDU	PDU	POW — Conversion, energy storage and distribution	POW
aidcf:RPP	RPP	POW — Conversion, energy storage and distribution	POW
aidcf:RackPDU	Rack PDU	POW — Conversion, energy storage and distribution	POW
aidcf:PowerSupplyUnit	Power Supply Unit	POW — Conversion, energy storage and distribution	POW, CMP
aidcf:ProtectionRelay	Protection Relay	POW — Protection, earthing and power quality	POW
aidcf:ProtectionDevice	Protection Device	POW — Protection, earthing and power quality	POW
aidcf:Relay	Relay	POW — Protection, earthing and power quality	POW
aidcf:EarthingSystem	Earthing System	POW — Protection, earthing and power quality	POW
aidcf:Grounding	Grounding	POW — Protection, earthing and power quality	POW
aidcf:LightningProtection	Lightning Protection	POW — Protection, earthing and power quality	POW, PHY

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:HarmonicFilter	Harmonic Filter	POW — Protection, earthing and power quality	POW
aidcf:SurgeProtection	Surge Protection	POW — Protection, earthing and power quality	POW
aidcf:EPMS	EPMS	POW — Electrical monitoring	POW, OPS, SEC
aidcf:Meter	Meter	POW — Electrical monitoring	POW
aidcf:PowerQualityMonitor	Power Quality Monitor	POW — Electrical monitoring	POW
aidcf:ChilledWaterPlant	Chilled Water Plant	THM — Air-side and mechanical systems	THM
aidcf:CondenserSystem	Condenser System	THM — Air-side and mechanical systems	THM
aidcf:PumpSystem	Pump System	THM — Air-side and mechanical systems	THM
aidcf:AirDistribution	Air Distribution	THM — Air-side and mechanical systems	THM
aidcf:Ventilation	Ventilation	THM — Air-side and mechanical systems	THM, ENV
aidcf:Pressurisation	Pressurisation	THM — Air-side and mechanical systems	THM
aidcf:AirHandler	Air Handler	THM — Air-side and mechanical systems	THM
aidcf:InRowCooler	In-Row Cooler	THM — Air-side and mechanical systems	THM
aidcf:RearDoorHeatExchanger	Rear-Door Heat Exchanger	THM — Air-side and mechanical systems	THM
aidcf:HeatSink	Heat Sink	THM — Heat capture at the IT load	THM, CMP
aidcf:ColdPlate	Cold Plate	THM — Heat capture at the IT load	THM, CMP
aidcf:ImmersionTank	Immersion Tank	THM — Heat capture at the IT load	THM, CMP
aidcf:CDU	CDU	THM — Coolant distribution	THM
aidcf:PrimaryLoop	Primary Loop	THM — Coolant distribution	THM
aidcf:SecondaryLoop	Secondary Loop	THM — Coolant distribution	THM
aidcf:Pump	Pump	THM — Coolant distribution	THM
aidcf:Manifold	Manifold	THM — Coolant distribution	THM
aidcf:QuickDisconnect	Quick Disconnect	THM — Coolant distribution	THM
aidcf:Hose	Hose	THM — Coolant distribution	THM
aidcf:Pipe	Pipe	THM — Coolant distribution	THM
aidcf:HeatExchanger	Heat Exchanger	THM — Coolant distribution	THM
aidcf:Chiller	Chiller	THM — Heat rejection	THM
aidcf:CoolingTower	Cooling Tower	THM — Heat rejection	THM, ENV
aidcf:DryCooler	Dry Cooler	THM — Heat rejection	THM
aidcf:Condenser	Condenser	THM — Heat rejection	THM
aidcf:EvaporativeSystem	Evaporative System	THM — Heat rejection	THM, ENV
aidcf:HeatReuseInterface	Heat-Reuse Interface	THM — Heat rejection	THM, ENV
aidcf:WaterSource	Water Source	THM — Fluid, water and leak management	THM, ENV, DEP
aidcf:Pipework	Pipework	THM — Fluid, water and leak management	THM
aidcf:Valve	Valve	THM — Fluid, water and leak management	THM
aidcf:Drain	Drain	THM — Fluid, water and leak management	THM
aidcf:Sump	Sump	THM — Fluid, water and leak management	THM
aidcf:Drainage	Drainage	THM — Fluid, water and leak management	THM
aidcf:LeakDetectionZone	Leak Detection Zone	THM — Fluid, water and leak management	THM
aidcf:WaterTreatmentSystem	Water Treatment System	THM — Fluid, water and leak management	THM, ENV
aidcf:TemperatureSensor	Temperature Sensor	THM — Thermal and leak monitoring	THM
aidcf:HumiditySensor	Humidity Sensor	THM — Thermal and leak monitoring	THM, ENV
aidcf:FlowSensor	Flow Sensor	THM — Thermal and leak monitoring	THM
aidcf:PressureSensor	Pressure Sensor	THM — Thermal and leak monitoring	THM
aidcf:LeakSensor	Leak Sensor	THM — Thermal and leak monitoring	THM
aidcf:CoolantQualitySensor	Coolant Quality Sensor	THM — Thermal and leak monitoring	THM
aidcf:ThermalCamera	Thermal Camera	THM — Thermal and leak monitoring	THM
aidcf:GridElectricity	Grid Electricity	ENV — Energy resources	ENV, POW
aidcf:RenewableElectricity	Renewable Electricity	ENV — Energy resources	ENV, POW
aidcf:OnSiteEnergy	On-Site Energy	ENV — Energy resources	ENV, POW
aidcf:StoredEnergy	Stored Energy	ENV — Energy resources	ENV, POW
aidcf:Fuel	Fuel	ENV — Energy resources	ENV, POW
aidcf:PotableWater	Potable Water	ENV — Water resources	ENV
aidcf:ProcessWater	Process Water	ENV — Water resources	ENV, THM
aidcf:ReclaimedWater	Reclaimed Water	ENV — Water resources	ENV
aidcf:CoolingWater	Cooling Water	ENV — Water resources	ENV, THM
aidcf:DischargeWater	Discharge Water	ENV — Water resources	ENV

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:AmbientTemperature	Ambient Temperature	ENV — Environmental conditions	ENV, THM
aidcf:Humidity	Humidity	ENV — Environmental conditions	ENV, THM
aidcf:AirQuality	Air Quality	ENV — Environmental conditions	ENV
aidcf:ParticulateLevel	Particulate Level	ENV — Environmental conditions	ENV
aidcf:Noise	Noise	ENV — Environmental conditions	ENV
aidcf:Vibration	Vibration	ENV — Environmental conditions	ENV, PHY
aidcf:FloodExposure	Flood Exposure	ENV — Environmental conditions	ENV, PHY
aidcf:HeatStress	Heat Stress	ENV — Environmental conditions	ENV, THM
aidcf:Refrigerant	Refrigerant	ENV — Material and chemical resources	ENV, THM
aidcf:Coolant	Coolant	ENV — Material and chemical resources	ENV, THM
aidcf:FireSuppressant	Fire Suppressant	ENV — Material and chemical resources	ENV, PHY
aidcf:BatteryChemistry	Battery Chemistry	ENV — Material and chemical resources	ENV, POW, PHY
aidcf:HazardousMaterial	Hazardous Material	ENV — Material and chemical resources	ENV, PHY
aidcf:PUERecord	PUE Record	ENV — Performance indicators	ENV
aidcf:WUERecord	WUE Record	ENV — Performance indicators	ENV
aidcf:CUERecord	CUE Record	ENV — Performance indicators	ENV
aidcf:EnergyReuseRecord	Energy Reuse Record	ENV — Performance indicators	ENV
aidcf:CoolingEfficiencyRecord	Cooling Efficiency Record	ENV — Performance indicators	ENV, THM
aidcf:ComputeUtilisationRecord	Compute Utilisation Record	ENV — Performance indicators	ENV, CMP
aidcf:Permit	Permit	ENV — Environmental obligations	ENV, GOV
aidcf:ReportingRequirement	Reporting Requirement	ENV — Environmental obligations	ENV, GOV
aidcf:EmissionLimit	Emission Limit	ENV — Environmental obligations	ENV
aidcf:WaterLimit	Water Limit	ENV — Environmental obligations	ENV
aidcf:WasteRequirement	Waste Requirement	ENV — Environmental obligations	ENV
aidcf:EnvironmentalCommitment	Environmental Commitment	ENV — Environmental obligations	ENV, GOV
aidcf:Accelerator	Accelerator	CMP — Accelerated compute	CMP
aidcf:GPU	GPU	CMP — Accelerated compute	CMP
aidcf:AIASIC	AI ASIC	CMP — Accelerated compute	CMP
aidcf:ComputeNode	Compute Node	CMP — Accelerated compute	CMP
aidcf:GPUBaseboard	GPU Baseboard	CMP — Accelerated compute	CMP
aidcf:CPU	CPU	CMP — Accelerated compute	CMP
aidcf:Memory	Memory	CMP — Accelerated compute	CMP
aidcf:DPU	DPU	CMP — Accelerated compute	CMP, NET
aidcf:NIC	NIC	CMP — Accelerated compute	CMP, NET
aidcf:ComputeTray	Compute Tray	CMP — Rack-scale systems	CMP
aidcf:RackScaleSystem	Rack-Scale System	CMP — Rack-scale systems	CMP, PHY, POW, THM
aidcf:ScaleUpDomain	Scale-Up Domain	CMP — Rack-scale systems	CMP, NET
aidcf:NVLinkDomain	NVLink Domain	CMP — Rack-scale systems	CMP, NET
aidcf:Enclosure	Enclosure	CMP — Rack-scale systems	CMP
aidcf:Backplane	Backplane	CMP — Rack-scale systems	CMP
aidcf:PowerShelf	Power Shelf	CMP — Rack-scale systems	CMP, POW
aidcf:RackController	Rack Controller	CMP — Rack-scale systems	CMP, SEC
aidcf:ComputeCluster	Compute Cluster	CMP — Cluster systems	CMP
aidcf:ClusterPartition	Cluster Partition	CMP — Cluster systems	CMP
aidcf:NodePool	Node Pool	CMP — Cluster systems	CMP
aidcf:HeadNode	Head Node	CMP — Cluster systems	CMP
aidcf:LoginNode	Login Node	CMP — Cluster systems	CMP, SEC
aidcf:ManagementNode	Management Node	CMP — Cluster systems	CMP, SEC
aidcf:ProvisioningNode	Provisioning Node	CMP — Cluster systems	CMP
aidcf:BMC	BMC	CMP — Management and firmware	CMP, SEC
aidcf:Firmware	Firmware	CMP — Management and firmware	CMP, SUP, SEC
aidcf:BIOS	BIOS	CMP — Management and firmware	CMP, SEC
aidcf:Driver	Driver	CMP — Management and firmware	CMP, SUP
aidcf:ClusterManager	Cluster Manager	CMP — Management and firmware	CMP, SEC
aidcf:ProvisioningService	Provisioning Service	CMP — Management and firmware	CMP
aidcf:Scheduler	Scheduler	CMP — Management and firmware	CMP
aidcf:Orchestrator	Orchestrator	CMP — Management and firmware	CMP

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:Queue	Queue	CMP — Workload infrastructure	CMP
aidcf:SchedulerPartition	Scheduler Partition †	CMP — Workload infrastructure	CMP
aidcf:Reservation	Reservation	CMP — Workload infrastructure	CMP
aidcf:ContainerRuntime	Container Runtime	CMP — Workload infrastructure	CMP
aidcf:KubernetesCluster	Kubernetes Cluster	CMP — Workload infrastructure	CMP
aidcf:JobScheduler	Job Scheduler	CMP — Workload infrastructure	CMP
aidcf:CheckpointAgent	Checkpoint Agent	CMP — Workload infrastructure	CMP, STO
aidcf:GPUTelemetry	GPU Telemetry	CMP — Compute observability	CMP
aidcf:NodeTelemetry	Node Telemetry	CMP — Compute observability	CMP
aidcf:HealthService	Health Service	CMP — Compute observability	CMP
aidcf:PerformanceCounter	Performance Counter	CMP — Compute observability	CMP
aidcf:ErrorCounter	Error Counter	CMP — Compute observability	CMP
aidcf:GoldenImage	Golden Image	CMP — Compute lifecycle	CMP, RES
aidcf:FirmwareBaseline	Firmware Baseline	CMP — Compute lifecycle	CMP, SUP
aidcf:CompatibilityMatrix	Compatibility Matrix	CMP — Compute lifecycle	CMP
aidcf:HardwareSupportContract	Hardware Support Contract	CMP — Compute lifecycle	CMP, SUP
aidcf:RMAProcess	RMA Process	CMP — Compute lifecycle	CMP, SUP
aidcf:SecureDecommissioningRecord	Secure Decommissioning Record	CMP — Compute lifecycle	CMP, PHY, EVA
aidcf:ParallelFileSystem	Parallel File System	STO — Storage systems	STO
aidcf:ScaleOutNAS	Scale-Out NAS	STO — Storage systems	STO
aidcf:ObjectStorage	Object Storage	STO — Storage systems	STO
aidcf:BlockStorage	Block Storage	STO — Storage systems	STO
aidcf:LocalNVMe	Local NVMe	STO — Storage systems	STO, CMP
aidcf:BurstBuffer	Burst Buffer	STO — Storage systems	STO
aidcf:ArchiveStorage	Archive Storage	STO — Storage systems	STO
aidcf:StorageNode	Storage Node	STO — Storage nodes and media	STO
aidcf:MetadataServer	Metadata Server	STO — Storage nodes and media	STO
aidcf:ObjectStorageNode	Object Storage Node	STO — Storage nodes and media	STO
aidcf:Controller	Controller	STO — Storage nodes and media	STO
aidcf:SSD	SSD	STO — Storage nodes and media	STO
aidcf:HDD	HDD	STO — Storage nodes and media	STO
aidcf:Tape	Tape	STO — Storage nodes and media	STO
aidcf:NVRAM	NVRAM	STO — Storage nodes and media	STO
aidcf:Namespace	Namespace	STO — Data-path services	STO
aidcf:MetadataService	Metadata Service	STO — Data-path services	STO
aidcf:DataMover	Data Mover	STO — Data-path services	STO
aidcf:Cache	Cache	STO — Data-path services	STO
aidcf:DataPathGateway	Data-Path Gateway †	STO — Data-path services	STO
aidcf:ClientDriver	Client Driver	STO — Data-path services	STO
aidcf:MountService	Mount Service	STO — Data-path services	STO
aidcf:DatasetRepository	Dataset Repository	STO — AI artifact services	STO
aidcf:ModelRepository	Model Repository	STO — AI artifact services	STO
aidcf:CheckpointRepository	Checkpoint Repository	STO — AI artifact services	STO, RES
aidcf:FeatureStore	Feature Store	STO — AI artifact services	STO
aidcf:ArtifactRegistry	Artifact Registry	STO — AI artifact services	STO
aidcf:Snapshot	Snapshot	STO — Protection and recovery	STO, RES
aidcf:Replica	Replica	STO — Protection and recovery	STO, RES
aidcf:BackupCopy	Backup Copy	STO — Protection and recovery	STO, RES
aidcf:ErasureCodingGroup	Erasure Coding Group	STO — Protection and recovery	STO
aidcf:RecoveryCopy	Recovery Copy	STO — Protection and recovery	STO, RES
aidcf:ImmutableCopy	Immutable Copy	STO — Protection and recovery	STO, RES, SEC
aidcf:StorageManager	Storage Manager	STO — Storage management	STO, SEC
aidcf:CapacityManager	Capacity Manager	STO — Storage management	STO
aidcf:TieringPolicy	Tiering Policy	STO — Storage management	STO
aidcf:RetentionPolicy	Retention Policy	STO — Storage management	STO, EVA
aidcf:IntegrityScanner	Integrity Scanner	STO — Storage management	STO
aidcf:TelemetryService	Telemetry Service	STO — Storage management	STO

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:IngestionInterface	Ingestion Interface	STO — Data logistics interfaces	STO
aidcf:ExportInterface	Export Interface	STO — Data logistics interfaces	STO
aidcf:DataTransferNode	Data Transfer Node	STO — Data logistics interfaces	STO, NET
aidcf:WANTransferService	WAN Transfer Service	STO — Data logistics interfaces	STO, NET, DEP
aidcf:DataStagingArea	Data Staging Area †	STO — Data logistics interfaces	STO
aidcf:ScaleUpFabric	Scale-Up Fabric	NET — Scale-up interconnect	NET, CMP
aidcf:SwitchFabric	Switch Fabric	NET — Scale-up interconnect	NET
aidcf:InterconnectLink	Interconnect Link	NET — Scale-up interconnect	NET
aidcf:TopologyDomain	Topology Domain	NET — Scale-up interconnect	NET
aidcf:InfiniBandFabric	InfiniBand Fabric	NET — Scale-out compute fabric	NET
aidcf:EthernetAIFabric	Ethernet AI Fabric	NET — Scale-out compute fabric	NET
aidcf:RoCEFabric	RoCE Fabric	NET — Scale-out compute fabric	NET
aidcf:ComputeLeaf	Compute Leaf	NET — Scale-out compute fabric	NET
aidcf:ComputeSpine	Compute Spine	NET — Scale-out compute fabric	NET
aidcf:FabricManager	Fabric Manager	NET — Scale-out compute fabric	NET, CMP, SEC
aidcf:StorageNetwork	Storage Network	NET — Storage fabric	NET, STO
aidcf:StorageLeaf	Storage Leaf	NET — Storage fabric	NET, STO
aidcf:StorageSpine	Storage Spine	NET — Storage fabric	NET, STO
aidcf:SANFabric	SAN Fabric	NET — Storage fabric	NET, STO
aidcf:StorageGateway	Storage Gateway	NET — Storage fabric	NET, STO
aidcf:FrontendNetwork	Frontend Network	NET — Frontend and service network	NET
aidcf:LoadBalancer	Load Balancer	NET — Frontend and service network	NET
aidcf:ServiceGateway	Service Gateway	NET — Frontend and service network	NET
aidcf:EdgeRouter	Edge Router	NET — Frontend and service network	NET, DEP
aidcf:Firewall	Firewall	NET — Frontend and service network	NET, SEC
aidcf:APINetwork	API Network	NET — Frontend and service network	NET
aidcf:ManagementNetwork	Management Network	NET — Management and control network	NET, SEC
aidcf:OutOfBandNetwork	Out-of-Band Network	NET — Management and control network	NET, SEC, RES
aidcf:ConsoleServer	Console Server	NET — Management and control network	NET, SEC
aidcf:ManagementSwitch	Management Switch	NET — Management and control network	NET
aidcf:JumpService	Jump Service	NET — Management and control network	NET, SEC
aidcf:TimeService	Time Service	NET — Timing and core services	NET
aidcf:DNSService	DNS Service	NET — Timing and core services	NET
aidcf:DHCPService	DHCP Service	NET — Timing and core services	NET
aidcf:IPAMService	IPAM Service	NET — Timing and core services	NET
aidcf:CertificateService	Certificate Service	NET — Timing and core services	NET, SEC
aidcf:AuthenticationService	Authentication Service	NET — Timing and core services	NET, SEC
aidcf:Cable	Cable	NET — Physical connectivity	NET, PHY
aidcf:OpticalFibre	Optical Fibre	NET — Physical connectivity	NET
aidcf:CopperLink	Copper Link	NET — Physical connectivity	NET
aidcf:Transceiver	Transceiver	NET — Physical connectivity	NET
aidcf:PatchPanel	Patch Panel	NET — Physical connectivity	NET
aidcf:MeetMeRoom	Meet-Me Room	NET — Physical connectivity	NET, PHY, DEP
aidcf:CrossConnect	Cross-Connect	NET — Physical connectivity	NET, DEP
aidcf:FlowTelemetry	Flow Telemetry	NET — Network observability	NET
aidcf:FabricTelemetry	Fabric Telemetry	NET — Network observability	NET
aidcf:PacketCaptureService	Packet Capture Service	NET — Network observability	NET, SEC
aidcf:NetworkController	Network Controller	NET — Network observability	NET, SEC
aidcf:CongestionMonitor	Congestion Monitor	NET — Network observability	NET
aidcf:PathMonitor	Path Monitor	NET — Network observability	NET
aidcf:Procedure	Procedure	OPS — Work control	OPS
aidcf:MethodStatement	Method Statement	OPS — Work control	OPS
aidcf:PermitToWork	Permit to Work	OPS — Work control	OPS, PHY
aidcf:LockoutTagout	Lockout Tagout	OPS — Work control	OPS, POW, PHY
aidcf:MaintenanceWindow	Maintenance Window	OPS — Work control	OPS
aidcf:ChangeRecord	Change Record	OPS — Work control	OPS
aidcf:TemporaryConfiguration	Temporary Configuration	OPS — Work control	OPS

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:MaintenancePlan	Maintenance Plan	OPS — Maintenance and reliability	OPS
aidcf:WorkOrder	Work Order	OPS — Maintenance and reliability	OPS
aidcf:ConditionMonitoring	Condition Monitoring	OPS — Maintenance and reliability	OPS
aidcf:Inspection	Inspection	OPS — Maintenance and reliability	OPS
aidcf:Calibration	Calibration	OPS — Maintenance and reliability	OPS, EVA
aidcf:SparePart	Spare Part	OPS — Maintenance and reliability	OPS, SUP
aidcf:ObsolescenceRecord	Obsolescence Record	OPS — Maintenance and reliability	OPS, SUP
aidcf:Shift	Shift	OPS — Staffing and competence	OPS
aidcf:Roster	Roster	OPS — Staffing and competence	OPS
aidcf:CompetencyRecord	Competency Record	OPS — Staffing and competence	OPS, GOV
aidcf:Authorisation	Authorisation	OPS — Staffing and competence	OPS
aidcf:TrainingRecord	Training Record	OPS — Staffing and competence	OPS
aidcf:Contractor	Contractor	OPS — Staffing and competence	OPS, SUP
aidcf:SubjectMatterExpert	Subject Matter Expert	OPS — Staffing and competence	OPS
aidcf:Alarm	Alarm	OPS — Alarm and incident	OPS
aidcf:Incident	Incident	OPS — Alarm and incident	OPS, RES
aidcf:EventLog	Event Log	OPS — Alarm and incident	OPS, CMP, SEC, EVA
aidcf:PostIncidentReview	Post-Incident Review	OPS — Alarm and incident	OPS, RES
aidcf:ConfigurationBaseline	Configuration Baseline	OPS — Configuration and documentation	OPS, SEC
aidcf:AssetRegister	Asset Register	OPS — Configuration and documentation	OPS, DEP
aidcf:DrawingRegister	Drawing Register	OPS — Configuration and documentation	OPS
aidcf:ProcedureRegister	Procedure Register	OPS — Configuration and documentation	OPS
aidcf:Revision	Revision	OPS — Configuration and documentation	OPS
aidcf:DocumentCustodian	Document Custodian	OPS — Configuration and documentation	OPS, EVA
aidcf:BMS	BMS	OPS — Facility monitoring and control	OPS, THM, SEC
aidcf:SCADA	SCADA	OPS — Facility monitoring and control	OPS, POW, THM, SEC
aidcf:PLC	PLC	OPS — Facility monitoring and control	OPS, THM, POW, SEC
aidcf:RTU	RTU	OPS — Facility monitoring and control	OPS, POW, SEC
aidcf:Sensor	Sensor	OPS — Facility monitoring and control	OPS, THM, POW
aidcf:Actuator	Actuator	OPS — Facility monitoring and control	OPS, THM, POW
aidcf:ControlSystemGateway	Control-System Gateway †	OPS — Facility monitoring and control	OPS, SEC, NET
aidcf:Historian	Historian	OPS — Facility monitoring and control	OPS, EVA, SEC
aidcf:AlarmConsole	Alarm Console	OPS — Facility monitoring and control	OPS
aidcf:EmergencyProcedure	Emergency Procedure	RES — Continuity and emergency response	RES, OPS
aidcf:CrisisTeam	Crisis Team	RES — Continuity and emergency response	RES, GOV
aidcf:Exercise	Exercise	RES — Continuity and emergency response	RES, EVA
aidcf:ContinuityPlan	Continuity Plan †	RES — Continuity and emergency response	RES
aidcf:RecoveryPlan	Recovery Plan †	RES — Continuity and emergency response	RES
aidcf:AlternativeCapacity	Alternative Capacity †	RES — Continuity and emergency response	RES, CAP, DEP
aidcf:Supplier	Supplier	SUP — Supplier and service management	SUP, DEP
aidcf:MaintenanceContract	Maintenance Contract	SUP — Supplier and service management	SUP, OPS
aidcf:ServiceLevel	Service Level	SUP — Supplier and service management	SUP, DEP
aidcf:RemoteSupport	Remote Support	SUP — Supplier and service management	SUP, SEC
aidcf:EscalationContact	Escalation Contact	SUP — Supplier and service management	SUP, OPS
aidcf:LeadTime	Lead Time	SUP — Supplier and service management	SUP
aidcf:SupplyConstraint	Supply Constraint	SUP — Supplier and service management	SUP, DEP
aidcf:SoftwareRelease	Software Release †	SUP — Provenance and software supply	SUP
aidcf:SignatureRecord	Signature Record †	SUP — Provenance and software supply	SUP, SEC
aidcf:SoftwareBillOfMaterials	Software Bill of Materials †	SUP — Provenance and software supply	SUP, SEC
aidcf:ComponentProvenanceRecord	Component Provenance Record †	SUP — Provenance and software supply	SUP, EVA
aidcf:IdentityProvider	Identity Provider †	SEC — Identity and privileged access	SEC
aidcf:HumanIdentity	Human Identity †	SEC — Identity and privileged access	SEC
aidcf:MachineIdentity	Machine Identity †	SEC — Identity and privileged access	SEC
aidcf:PrivilegedAccessService	Privileged Access Service †	SEC — Identity and privileged access	SEC
aidcf:BreakGlassAccount	Break-Glass Account †	SEC — Identity and privileged access	SEC, RES
aidcf:RemoteAccessGateway	Remote Access Gateway †	SEC — Identity and privileged access	SEC, NET, SUP
aidcf:SecurityTelemetryService	Security Telemetry Service †	SEC — Security monitoring and response	SEC, EVA

Canonical ID	Preferred label	Part V family	Domain code(s)
aidcf:SecurityIncident	Security Incident †	SEC — Security monitoring and response	SEC, OPS, RES
aidcf:VulnerabilityRecord	Vulnerability Record †	SEC — Security monitoring and response	SEC, SUP
aidcf:AutomationAgent	Automation Agent †	SEC — Automated and AI-enabled action	SEC, OPS
aidcf:CommandAllowlist	Command Allowlist †	SEC — Automated and AI-enabled action	SEC
aidcf:ApprovalWorkflow	Approval Workflow †	SEC — Automated and AI-enabled action	SEC, GOV

A.3 Assurance-chain classes

Defined in Part XV (§68). All carry the primary domain code EVA because they are governed by EVA-01 to EVA-06; Control Domain, Control Family and Control are additionally the organising classes of every domain.

Canonical ID	Preferred label	Domain code(s)	AIDCF-014 class
aidcf:ControlDomain	Control Domain	EVA (all domains)	ControlDomain
aidcf:ControlFamily	Control Family	EVA (all domains)	ControlFamily
aidcf:Control	Control	EVA (all domains)	Control
aidcf:EvidenceRequirement	Evidence Requirement	EVA	EvidenceRequirement
aidcf:EvidenceObject	Evidence Object	EVA	EvidenceObject
aidcf:EvidencePackage	Evidence Package	EVA	EvidencePackage
aidcf:Assertion	Assertion	EVA	— (assertionRefs; AIDCF-014 Assertion vocabulary)
aidcf:ControlAssessment	Control Assessment	EVA	ControlAssessment
aidcf:Finding	Finding	EVA	Finding
aidcf:UnknownRecord	UNKNOWN Record	EVA	UnknownRecord
aidcf:CriticalGate	Critical Gate	EVA, GOV	CriticalGate
aidcf:GateResult	Gate Result	EVA	GateResult
aidcf:ScoreSnapshot	Score Snapshot	EVA	ScoreSnapshot
aidcf:DecisionRecommendation	Decision Recommendation	EVA	DecisionRecommendation
aidcf:AssuranceDecision	Assurance Decision	EVA, GOV	AssuranceDecision
aidcf:AssuranceStatement	Assurance Statement	EVA	AssuranceStatement
aidcf:AssuranceReport	Assurance Report	EVA	AssuranceReport
aidcf:ReassessmentTrigger	Reassessment Trigger	EVA	ReassessmentTrigger
aidcf:DecisionQuestion	Decision Question	EVA, GOV	— (decisionQuestion attribute of Assessment and AssuranceDecision)

A.4 Industrial automation and control system security classes

Defined in Part XVI (§72).

Canonical ID	Preferred label	Domain code(s)	AIDCF-014 class
aidcf:IACS	Industrial Automation and Control System (IACS)	SEC, OPS, POW, THM	InfrastructureSystem (systemType = IACS)
aidcf:IACSZone	Zone (IEC 62443)	SEC	— (InfrastructureSystem grouping, systemType = IACS_ZONE)
aidcf:Conduit	Conduit (IEC 62443)	SEC, NET	Interface (interfaceType = IACS_CONDUIT)
aidcf:SecurityLevel	Security Level	SEC	— (attribute vector on Zone/Conduit/Component)
aidcf:FoundationalRequirement	Foundational Requirement	SEC	— (controlled vocabulary FR1–FR7)

A.5 Aliases and disambiguation

Term	Treatment
CriticalFunction (v1.0)	Replaced by aidcf:RequiredFunction.
AgentSystem (v1.0)	Renamed aidcf:SuppressionAgentSystem to avoid confusion with AI agents (aidcf:AutomationAgent).
Gateway (v1.0, two uses)	Split into aidcf:ControlSystemGateway (OPS/SEC) and aidcf:DataPathGateway (STO).
Partition (v1.0)	Renamed aidcf:SchedulerPartition; distinct from aidcf:ClusterPartition (hardware/cluster subdivision).
Staging Area	aidcf:StagingArea is a physical area (PHY); aidcf:DataStagingArea is a data-logistics location (STO).
Security Zone	aidcf:SecurityZone is a physical access zone (PHY); aidcf:IACSZone is an IEC 62443 zone (SEC). A physical zone may bound an IACS zone but is not one.
Transformer, Role, Event Log (v1.0 duplicates)	Each now registered once with multiple domain codes.
Grounding / Earthing System	aidcf:Grounding is retained as a regional synonym label of aidcf:EarthingSystem; implementations should emit EarthingSystem.
Relay / Protection Relay	aidcf:ProtectionRelay is a protective relay; aidcf:Relay is an auxiliary or control relay without a protection function.
NVLink Domain, Kubernetes Cluster	Product-specific specialisations of aidcf:ScaleUpDomain and aidcf:Orchestrator respectively (IOT-REQ-013).

Appendix B - Canonical relationship registry

Ontology predicates are listed with their logic and with the AIDCF-014 §9 relationship type used for exchange (see the exchange requirement in §34). Assurance-chain relationships are listed in §70.

Predicate	Subject	Object	Logic	Inverse	Meaning	AIDCF-014 exchange type
PART_OF	Entity	Entity	Transitive	HAS_PART	Composition.	COMPOSED_OF (inverse)
CONTAINS	Location/System	Entity	Transitive	LOCATED_IN / PART_OF	Containment.	COMPOSED_OF or LOCATED_IN (inverse)
LOCATED_IN	Physical Entity	Location	Transitive	CONTAINS	Physical placement.	LOCATED_IN
MEMBER_OF	Entity	Group	Not transitive	HAS_MEMBER	Group membership.	COMPOSED_OF (inverse, from the group)
HAS_OUTCOME	CapabilityInstance	ServiceOutcome	Not transitive	—	Capability requires outcome.	HAS_OUTCOME
REQUIRES_FUNCTION	ServiceOutcome	RequiredFunction	Not transitive	—	Outcome depends on function.	REQUIRES_FUNCTION
IMPLEMENTED_BY	RequiredFunction	System/Asset	Not transitive	IMPLEMENTS	Function realised by system or asset.	IMPLEMENTED_BY
INTERFACES_WITH	Entity	Entity	Symmetric	INTERFACES_WITH	Defined exchange boundary.	CONNECTS_TO via an Interface record
CONNECTS_TO	Entity	Entity	Symmetric unless directed	CONNECTS_TO	Realised connection.	CONNECTS_TO
ROUTES_THROUGH	Flow/Service	Entity	Transitive along a path	CARRIES	Path relationship.	DEPENDS_ON (dependencyType = Network, path qualifier)
DEPENDS_ON	Consumer entity	Provider/entity/condition	Directional; transitive for path tracing only	None registered	General reliance.	DEPENDS_ON
POWERED_BY	Load/system	Electrical source/path	Directional; transitive along the electrical path	None registered	Electrical reliance.	DEPENDS_ON (dependencyType = Electrical); CONSUMES_RESOURCE for energy
COOLED_BY	Heat source/system	Cooling system/path	Directional; transitive along the heat path	None registered	Thermal reliance.	DEPENDS_ON (dependencyType = Thermal)
SUPPLIED_BY	Consumer	Provider/resource	Directional	None registered	Resource or service supply.	CONSUMES_RESOURCE (resource) or DEPENDS_ON (service)
CONTROLLED_BY	Asset/system	Controller/actor	Directional	None registered	Operational or automated control.	CONTROLLED_BY
MONITORED_BY	Entity	Monitoring entity	Directional	None registered	Visibility and detection reliance.	MONITORED_BY
PROTECTED_BY	Entity	Safeguard	Directional	None registered	Protective relationship.	PROTECTED_BY
AUTHENTICATED_BY	Entity/service	Identity service	Directional	None registered	Identity reliance.	DEPENDS_ON (dependencyType = Identity)
TIMED_BY	Entity/service	Time source	Directional	None registered	Timing reliance.	DEPENDS_ON (dependencyType = Timing)
CONFIGURED_BY	Entity	Configuration source/actor	Directional	None registered	Configuration reliance.	DEPENDS_ON (dependencyType = Configuration)
RECOVERS_THROUGH	Entity/capability	Recovery asset/process	Directional	None registered	Restoration reliance.	RECOVERS_THROUGH
MAINTAINED_BY	Entity	Actor/organisation	Directional	None registered	Maintenance reliance.	OPERATED_BY (role = maintainer)
OPERATED_BY	Entity/system	Actor/organisation	Directional	None registered	Operational responsibility.	OPERATED_BY
EVIDENCED_BY	Assertion/entity	Evidence object	Directional	None registered	Evidence support.	EVIDENCED_BY
TESTED_BY	Assertion/safeguard	Test record	Directional	None registered	Validation link.	ASSESSED_BY (method = TEST)
SHARES_DEPENDENCY_WITH	Entity	Entity	Symmetric; derived	SHARES_DEPENDENCY_WITH	Common provider exposure.	Derived from two DEPENDS_ON to one provider; may be materialised as DEPENDS_ON pairs
SHARES_FAULT_DOMAIN_WITH	Entity	Entity	Symmetric	SHARES_FAULT_DOMAIN_WITH	Shared failure exposure.	SHARES_FAULT_DOMAIN_WITH

Predicate	Subject	Object	Logic	Inverse	Meaning	AIDCF-014 exchange type
DIVERSE_FROM	Entity	Entity	Symmetric; characteristic-specific	DIVERSE_FROM	Declared material difference.	Assertion on the RedundancyGroup (no AIDCF-014 type)
INDEPENDENT_OF	Entity	Entity	Symmetric; condition-bounded	INDEPENDENT_OF	Evidence-bounded relevant independence.	Assertion with EVIDENCED_BY (no AIDCF-014 type)
REDUNDANT_WITH	Entity	Entity	Symmetric within a RedundancyGroup	REDUNDANT_WITH	Alternative for a protected function.	COMPOSED_OF from the RedundancyGroup
CONCENTRATES	Entity	Entity	Directional	None registered	Aggregates capability or consequence.	Assertion on the Dependency (no AIDCF-014 type)
SINGLE_POINT_FOR	Entity	Function/Outcome	Directional	None registered	Unmitigated single-point relationship.	Finding or Assertion referencing DEPENDS_ON (no AIDCF-014 type)
ISOLATED_FROM	Entity	Entity	Symmetric; hazard- or threat-specific	ISOLATED_FROM	Declared separation.	Assertion with EVIDENCED_BY (no AIDCF-014 type)

Legacy predicates: SUPPORTS (entity to outcome) is withdrawn in favour of HAS_OUTCOME / REQUIRES_FUNCTION / IMPLEMENTED_BY; IMPLEMENTS is the inverse of IMPLEMENTED_BY; CONNECTED_TO is renamed CONNECTS_TO; RECOVERED_BY is renamed RECOVERS_THROUGH; IN_SAME_FAULT_DOMAIN_AS is renamed SHARES_FAULT_DOMAIN_WITH.

Appendix C - Minimum property schemas

C.1 Capability Instance

Property	Cardinality	Type
capabilityInstanceId	1	Identifier
name	1	String
capabilityType	1	Controlled term
owner	1	Party
lifecycleStage	1	Enumeration
serviceOutcomes	1..*	ServiceOutcome
criticality	1..*	Criticality
assessmentBoundary	1	Boundary
operatingEnvelope	1	Envelope
architectureBaseline	1..*	Document/Configuration
effectivePeriod	1	Temporal interval
evidenceStatus	1	Evidence status (§53)

C.2 Dependency

Property	Cardinality	Type
dependencyId	1	Identifier
consumer	1	Entity
provider	1	Entity or UNKNOWN
dependencyType	1	Controlled term
strength	1	Enumeration
applicableStates	1..*	OperatingState
capacityRequirement	0..*	CapacityAssertion
substitutability	0..1	Enumeration
owner	1	Party or UNKNOWN
evidence	0..*	EvidenceObject
evidenceStatus	1	Evidence status
confidence	0..1	CONF-0..CONF-4
validity	1	Temporal interval

C.3 Evidence Object

Property	Cardinality	Type
evidenceId	1	Identifier

Property	Cardinality	Type
title	1	String
evidenceClass	1	E1–E7 (AIDCF-008)
source	1	Source
custodian	1	Party
collectionMethod	1	Controlled term
observationTime	1	Date/time
effectivePeriod	0..1	Temporal interval
scope	1	Entity set
integrity	1	I0–I5 (AIDCF-008)
qualityBand	1	Q0–Q5 (AIDCF-008)
supportsOrChallenges	1..*	Assertion
limitations	0..*	String
reviewer	0..1	Party
evidenceStatus	1	Evidence status

Appendix D - Naming, identifier and namespace rules

- Use singular nouns for classes and active predicates for relationships.
- Avoid acronyms as canonical labels unless universally necessary; retain aliases separately.
- Record regional spellings as labels without creating duplicate semantic classes.
- Keep metrics and units separate from values.
- Do not encode confidentiality, location or owner in public identifiers.
- Use explicit version metadata for vocabularies, schemas and profiles.
- Mark deprecated terms with replacement mappings and retirement rationale.
- Maintain a collision register for extensions and imported vocabularies.

Appendix E - Standards and reference-source map

The sources below informed scope comparisons and terminology analysis. AIDCF does not reproduce proprietary requirements and does not claim conformity with a referenced standard merely by mapping concepts.

Reference	Publisher/type	Relevance
AIDCF-001 AIDCF Manifesto	Internal canonical source	Purpose, doctrine, fifteen-domain architecture, evidence and UNKNOWN principles.
AIDCF-002 Core Conceptual Model	Internal canonical source	Capability, assets, dependencies, capacity, failure, safeguard and assurance concepts; canonical chain; MATERIAL; legacy domain mapping.
AIDCF-007A Canonical Master Control Registry	Internal canonical source	Fifteen domain codes, 90 families and 360 controls to which Part V and Part XV align.
AIDCF-014 Traceability and Machine-Executable Data Standard	Internal canonical source	Exchange object classes (§7) and relationship types (§9) to which Appendices A and B map.
ISO/IEC/IEEE 15026-2	ISO/IEC/IEEE	Assurance-case structure (claims, arguments, evidence) underlying the canonical chain.
IEC 62443 series (in particular parts 1-1, 3-2, 3-3 and 4-2)	IEC	Industrial automation and control system security concepts: zones, conduits, foundational requirements and security levels (Part XVI).
W3C OWL 2, SHACL, JSON-LD 1.1 and PROV-O	W3C	Machine-readable publication formats (Part XIV).
ISO/IEC 22237-1:2021	ISO	General data-centre concepts, reference models and classification context; the standard explicitly excludes selection/configuration of IT and network equipment.
ISO/IEC 22237 series	ISO/IEC	Facilities and infrastructure topics spanning building, power, environmental control, cabling, security and operations.
EN 50600 series	CENELEC	European data-centre facilities and infrastructure planning, construction and operation context.
NVIDIA Enterprise Reference Architectures	NVIDIA	Validated and repeatable AI-factory designs integrating compute, networking, storage and software.
Open Systems for AI: Blueprint for Scalable Infrastructure	Open Compute Project	Open and interoperable AI infrastructure ecosystem context.
AI-Ready Data Storage Infrastructure taxonomy	IDC	Storage-specific AI readiness and data logistics taxonomy context.
Cisco AI-ready infrastructure design guides	Cisco	Examples of AI infrastructure design profiles and validated designs.

E.1 Public links

ISO/IEC 22237-1:2021: https://www.iso.org/standard/78550.html

NVIDIA Reference Architectures: https://docs.nvidia.com/reference-architectures/index.html

NVIDIA Enterprise Reference Architectures: https://www.nvidia.com/en-us/technologies/enterprise-reference-architecture/

Open Systems for AI whitepaper: https://www.opencompute.org/documents/ocp-open-systems-for-ai-whitepaper-v1-0-0-final-pdf/

Cisco AI-ready infrastructure design guides: https://www.cisco.com/c/en/us/solutions/design-zone/ai-ready-infrastructure.html

Appendix F - Review and acceptance record

Review role	Status	Acceptance focus
Methodology author	Draft completed	Internal coherence and alignment with AIDCF-001 and AIDCF-002.
Ontology/knowledge-model reviewer	Pending	Class, relationship, constraint and machine-readable integrity.
Data-centre architecture reviewer	Pending	Facility and system taxonomy usability.
Electrical engineering reviewer	Pending	Power semantics and dependency accuracy.
Mechanical/cooling reviewer	Pending	Thermal, hydraulic and liquid-cooling semantics.
Fire/life-safety reviewer	Pending	Boundary and safety terminology.
Physical-security reviewer	Pending	Physical control and zoning taxonomy.
AI compute/HPC reviewer	Pending	Compute, rack-scale and cluster taxonomy.
Storage reviewer	Pending	Data infrastructure and recovery semantics.
Network/interconnect reviewer	Pending	Scale-up, scale-out and management fabrics.
OT/security reviewer	Pending	BMS, EPMS, SCADA, IEC 62443 zone/conduit and control relationships.
Assurance/evidence reviewer	Pending	UNKNOWN, confidence, provenance and claims.
AI Trust Graph integration reviewer	Pending	Handoff and shared-node semantics.
Legal/IP/trademark review	Pending	Publication claims, licensing and ownership.
Public release	Not approved	All mandatory review gates closed.

No approval has been granted for this document; the record states review status only.

F.1 Acceptance criteria

- Canonical terms remain consistent with the Manifesto and Core Conceptual Model.
- All fifteen domains are represented without becoming isolated silos, and every class carries a valid domain code.
- Capability, outcome, function, system, asset and dependency distinctions are operationally usable.
- Capacity and redundancy claims are condition- and evidence-bounded.
- Time, state, responsibility and UNKNOWN are first-class semantics.
- AIDCF-004 can attach threats, hazards, failures and propagation paths without restructuring the ontology.
- The assurance half of the chain (Part XV) can be populated for a synthetic assessment and exchanged through AIDCF-014 without loss.
- Machine-readable schemas can be generated from the normative registries.
- No content creates an unsupported certification or engineering claim.
- Independent multidisciplinary review and change control are recorded.

Change record

Version	Status	Summary
0.1	Architecture draft	Established upper ontology and initial domain taxonomy.
0.5	Integrated draft	Added relationships, state, capacity, responsibility, validation and profiles.
1.0	Controlled-review draft	Baseline for expert review and machine-readable implementation.
1.1	Draft for controlled review	15-domain re-baseline; assurance-chain and IEC 62443 classes; AIDCF-014 alignment; renumbered requirements.

Final doctrine

AIDCF infrastructure is represented only when its meaning can be connected to a Capability Instance, Service Outcome, Required Function, dependency, state or assurance purpose. Classification alone is insufficient. The ontology must reveal how the system is assembled, what it relies upon, under which conditions claims apply, who is responsible, what evidence supports the representation and what remains UNKNOWN.

Appendix G - Legacy identifier mapping

Requirement identifiers were renumbered in v1.1 to remove gaps. The table maps v1.0 identifiers to v1.1 identifiers; requirements new in v1.1 are listed for completeness.

v1.0 ID	v1.1 ID
IOT-REQ-001	IOT-REQ-001

v1.0 ID	v1.1 ID
IOT-REQ-002	IOT-REQ-002
IOT-REQ-003	IOT-REQ-003
IOT-REQ-004	IOT-REQ-004
IOT-REQ-005	IOT-REQ-005
IOT-REQ-006	IOT-REQ-006
IOT-REQ-016	IOT-REQ-007
IOT-REQ-017	IOT-REQ-008
IOT-REQ-018	IOT-REQ-009
IOT-REQ-019	IOT-REQ-010
— (new in v1.1)	IOT-REQ-011
— (new in v1.1)	IOT-REQ-012
IOT-REQ-020	IOT-REQ-013
IOT-REQ-021	IOT-REQ-014
IOT-REQ-028	IOT-REQ-015
IOT-REQ-029	IOT-REQ-016
IOT-REQ-034	IOT-REQ-017
IOT-REQ-035	IOT-REQ-018
IOT-REQ-036	IOT-REQ-019
IOT-REQ-037	IOT-REQ-020
— (new in v1.1)	IOT-REQ-021
IOT-REQ-041	IOT-REQ-022
IOT-REQ-042	IOT-REQ-023
IOT-REQ-043	IOT-REQ-024
IOT-REQ-046	IOT-REQ-025
IOT-REQ-047	IOT-REQ-026
IOT-REQ-048	IOT-REQ-027
IOT-REQ-051	IOT-REQ-028
IOT-REQ-052	IOT-REQ-029
IOT-REQ-054	IOT-REQ-030
IOT-REQ-055	IOT-REQ-031
IOT-REQ-056	IOT-REQ-032
IOT-REQ-066	IOT-REQ-033
IOT-REQ-067	IOT-REQ-034
IOT-REQ-068	IOT-REQ-035
— (new in v1.1)	IOT-REQ-036
— (new in v1.1)	IOT-REQ-037
— (new in v1.1)	IOT-REQ-038
— (new in v1.1)	IOT-REQ-039
— (new in v1.1)	IOT-REQ-040
— (new in v1.1)	IOT-REQ-041
— (new in v1.1)	IOT-REQ-042
— (new in v1.1)	IOT-REQ-043
— (new in v1.1)	IOT-REQ-044
— (new in v1.1)	IOT-REQ-045
— (new in v1.1)	IOT-REQ-046

Validation rules VAL-001 to VAL-020 are renamed IOT-VAL-001 to IOT-VAL-020 with unchanged numbers (IOT-VAL-021 to IOT-VAL-025 are new). Conformance levels C0 to C4 are renamed IOT-CL0 to IOT-CL4. The v1.0 artifact identifier AIDCF-IOT-001 is replaced by the public ID AIDCF-003.